

Practical Lab Report — Network Security Essentials

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Platform: TryHackMe — Network Security Essentials

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Project: Implementing a Secure Network Architecture for a Large Enterprise

Organisation: Spinnaker Analytics

Room completed: 8 tasks · 96 points · User: sawant052

1. Overview

- This report documents the completion of the TryHackMe room 'Network Security Essentials' as the practical component of Perimeter Security of the Spinnaker Analytics project.
- The lab involved analysing real perimeter security logs, firewall logs, VPN authentication logs, WAF logs, IDS/IPS alerts and correlating findings using Splunk SIEM.0.
- .
- The lab directly validates the perimeter security controls documented in the Perimeter Security Configuration Guide, providing hands-on evidence of how NGFW, WAF, VPN gateway, and IDPS generate logs and how those logs are analysed to detect attacks.

2. Lab Environment

- Platform: TryHackMe browser-based Linux VM (Ubuntu)
- Tools used: Pluma text editor, bash terminal, Splunk 10.0.0 (local instance)
- Log files analysed: firewall_logs.txt, vpn_logs.txt, waf_logs.txt, ids_alerts.log, ids_alerts.json
- SIEM: Splunk with network_logs index ingesting all log sources

3. WAF Log Analysis — Attack Detection

3.1 SQL Injection Detection

- The WAF log was analysed to identify blocked attack attempts.
- Line 10 of waf_logs.txt showed a BLOCK action against a SQL injection attempt from IP 198.51.100.12, targeting /products.php with a UNION SELECT payload.
- The WAF rule_id 942100 triggered the block.

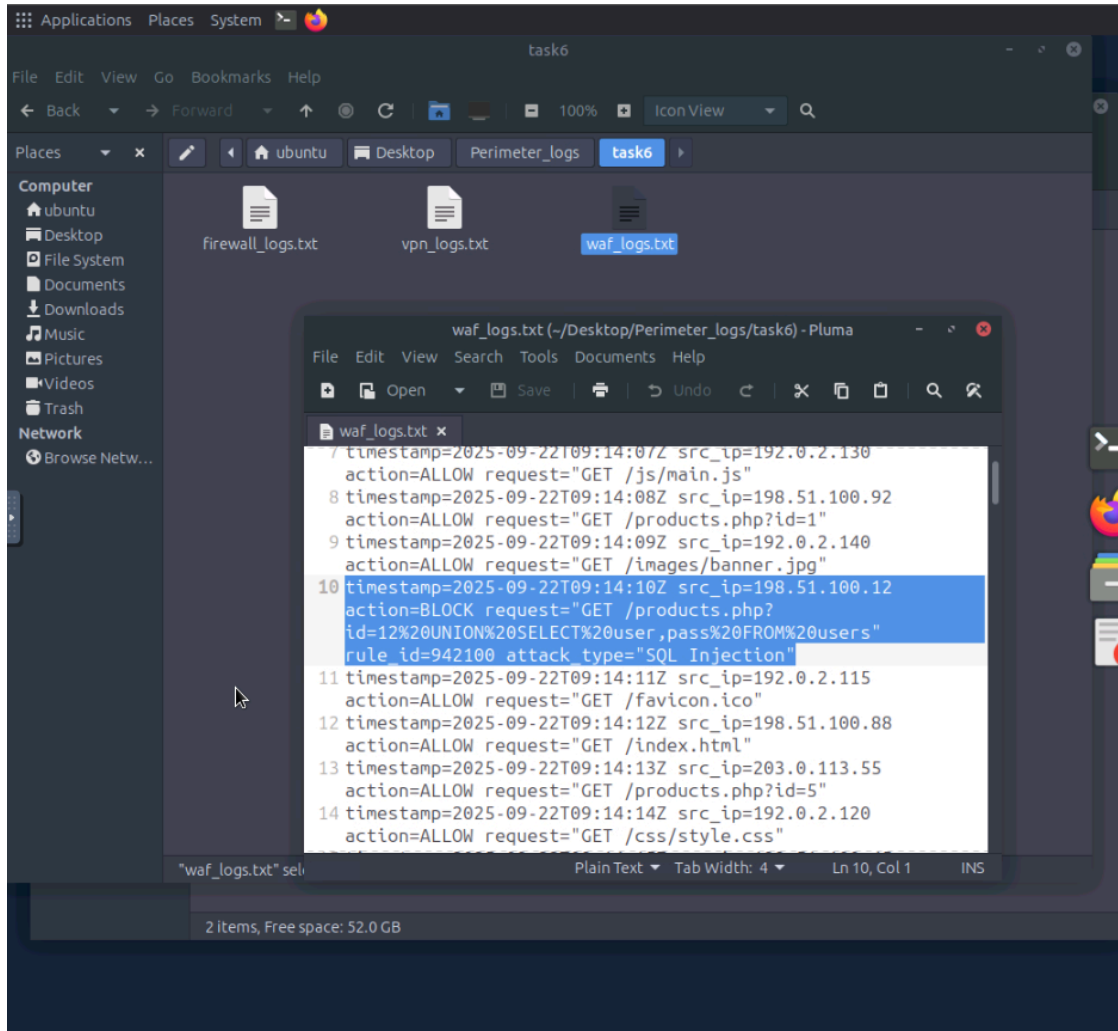


Figure 1: WAF log — SQL injection blocked (rule_id 942100, src: 198.51.100.12)

3.2 XSS Detection

- Further analysis of waf_logs.txt identified a cross-site scripting (XSS) attempt at line 46.
- The attacker at 198.51.100.12 attempted to inject a script tag via /search.php.
- The WAF rule_id 941100 blocked the request.

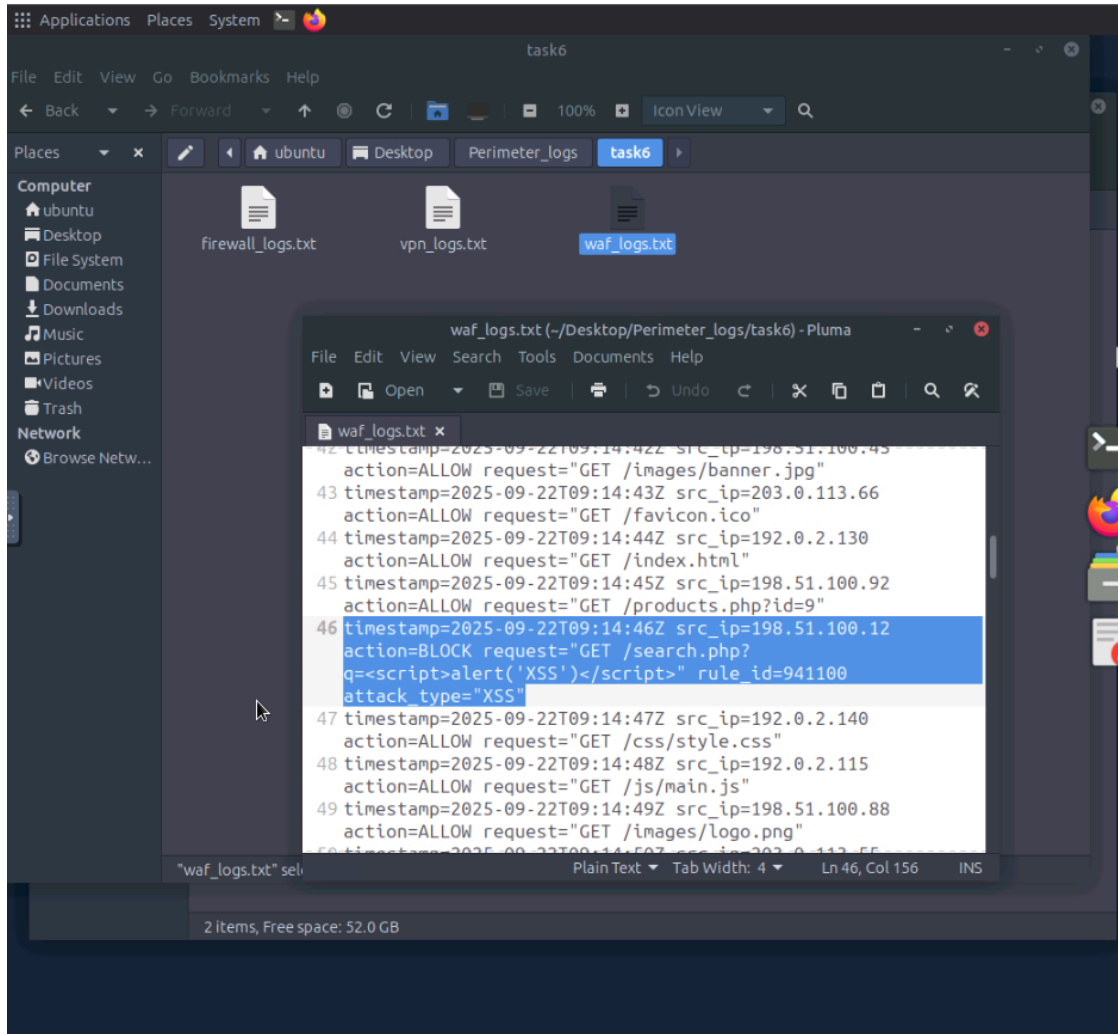


Figure 2: WAF log — XSS attack blocked (rule_id 941100, src: 198.51.100.12)

3.3 Directory Traversal Detection

- A directory traversal attack was identified at line 102 of waf_logs.txt.
- The attacker attempted to access ../../../../etc/passwd via a GET request.
- WAF rule_id 930120 blocked the request, preventing access to the system password file.

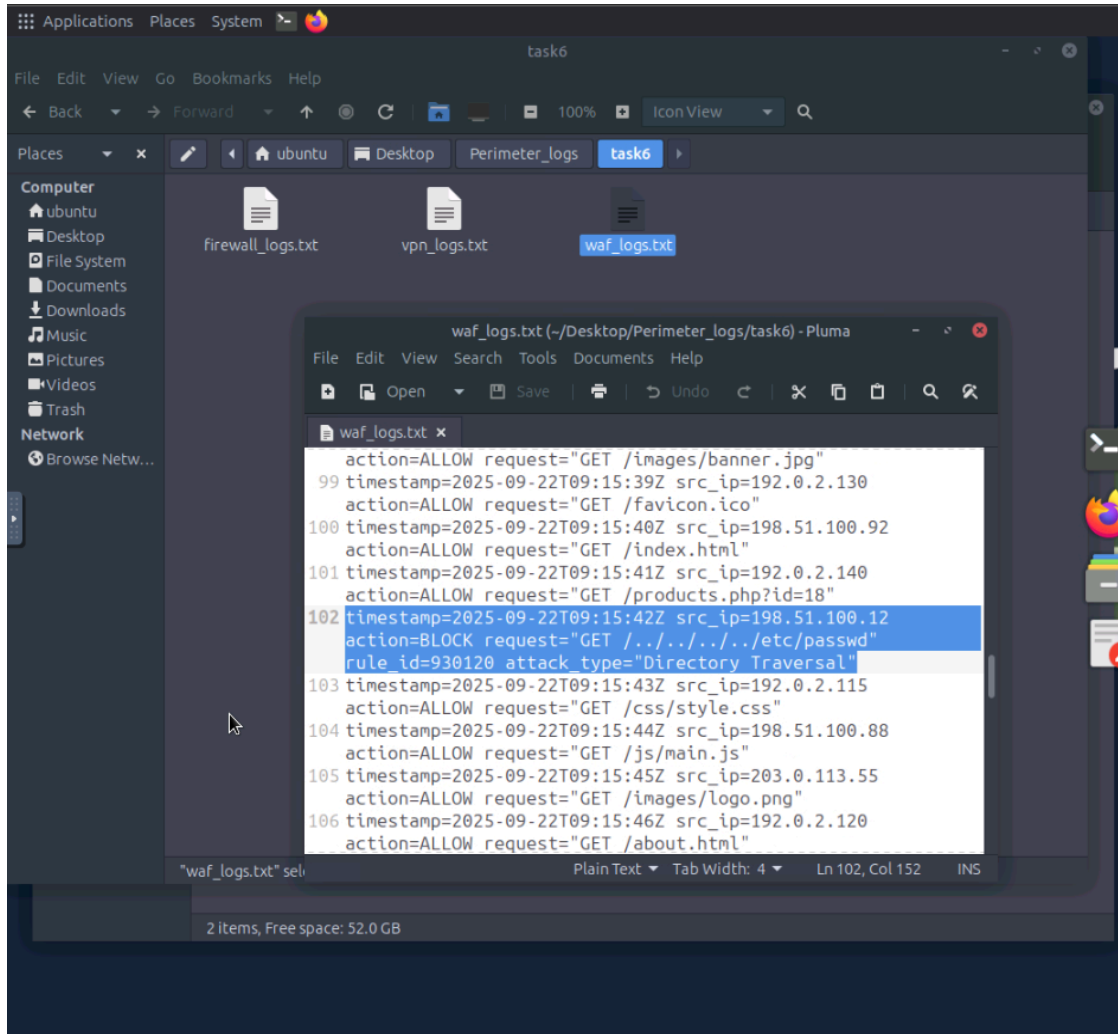


Figure 3: WAF log — Directory traversal blocked (rule_id 930120, src: 198.51.100.12)

3.4 Additional WAF Findings

- Screenshots 4 show further WAF log entries, confirming continued attack attempts from 198.51.100.12 across multiple attack vectors throughout the session window.

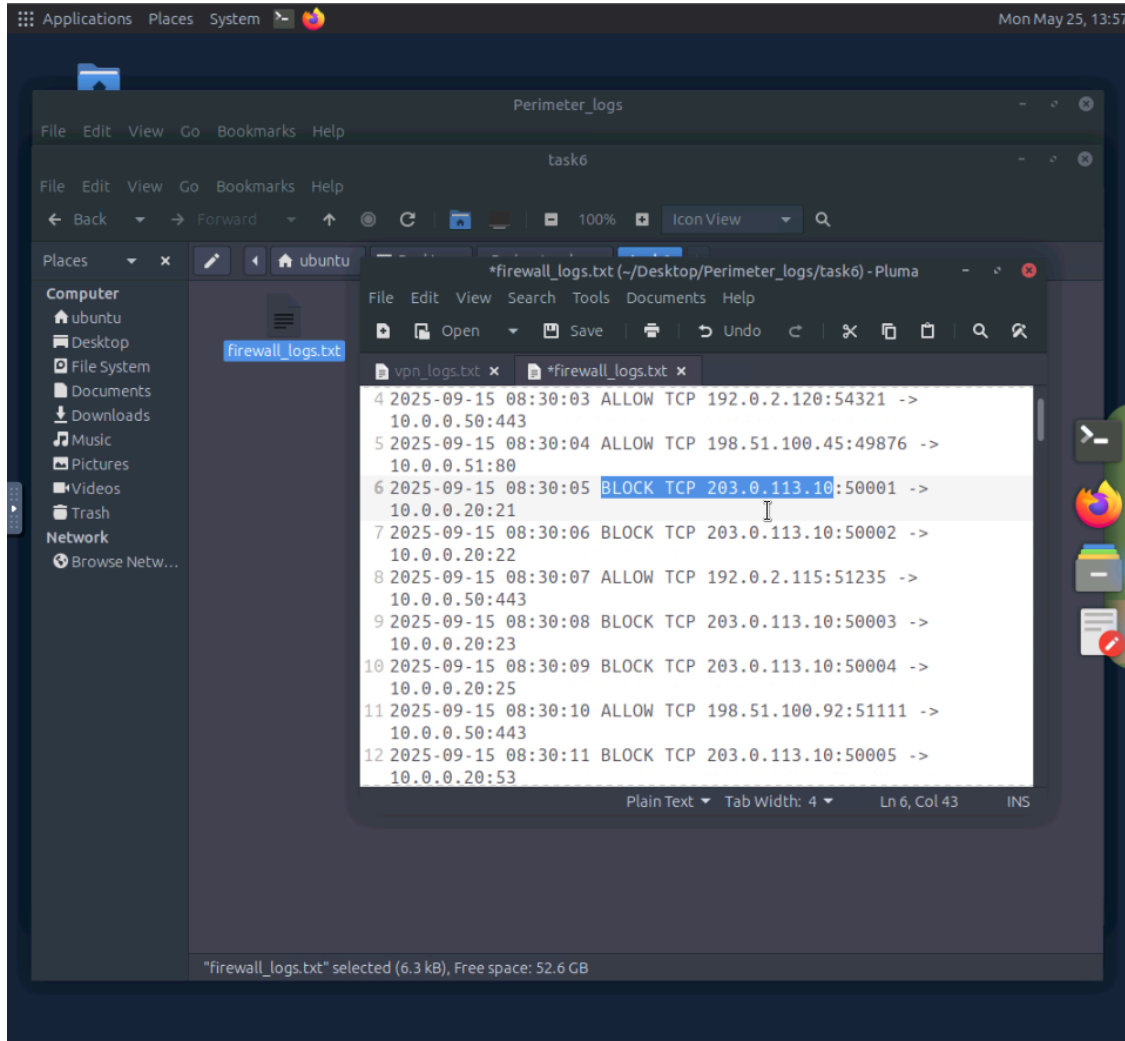


Figure 4: WAF log — continued attack traffic from 198.51.100.12

4. VPN Log Analysis — Brute-Force Detection

4.1 Credential Stuffing / Brute-Force Attack

- Analysis of vpn_logs.txt revealed a sustained credential stuffing attack originating from IP 45.137.22.13.
- The attacker attempted authentication with a long list of common usernames : root, test, guest, support, backup, user, ftpuser, webmaster, sales, info, demo, login, student, manager, hr, it, marketing, finance, billing, john, jane, susan, and many others ;
- All resulting in FAILED_AUTH responses.

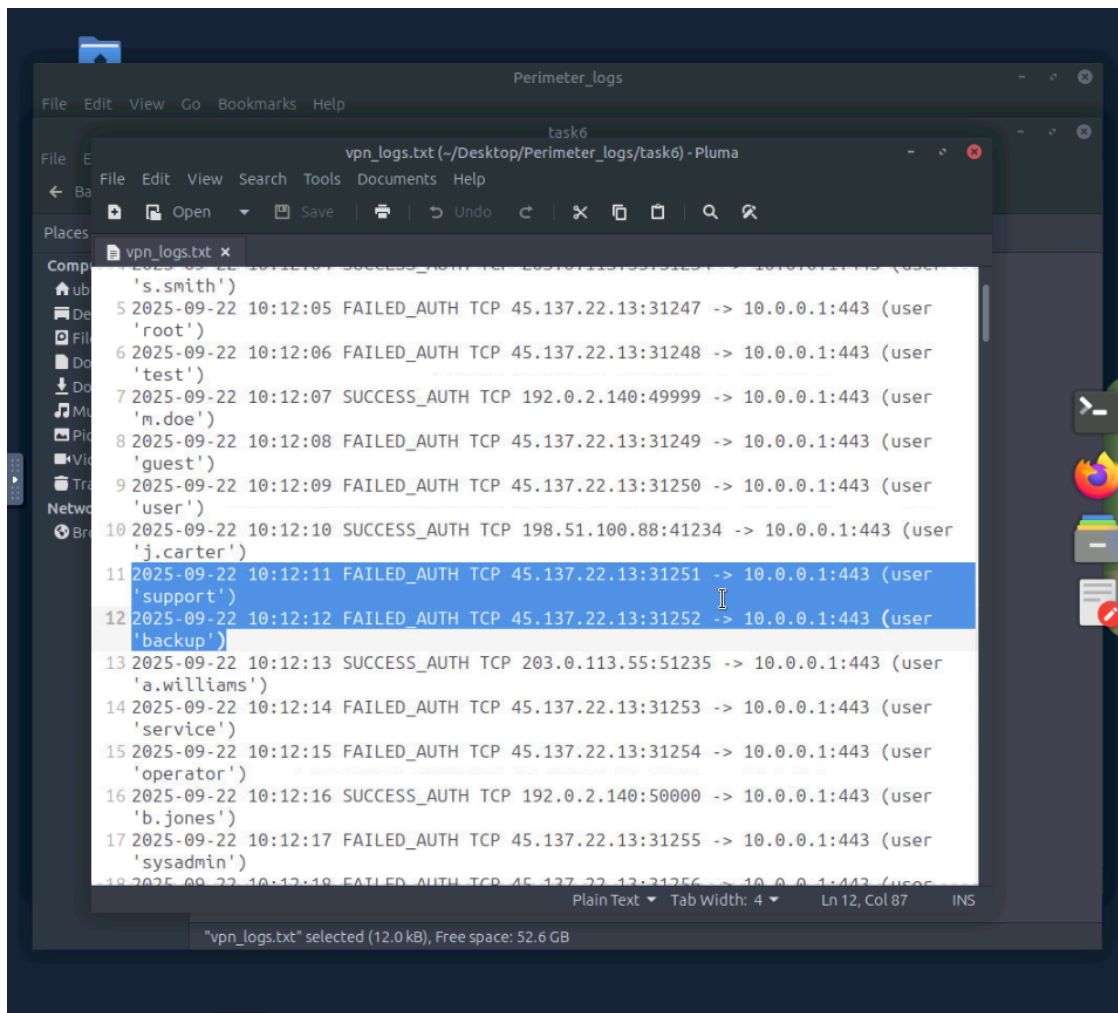


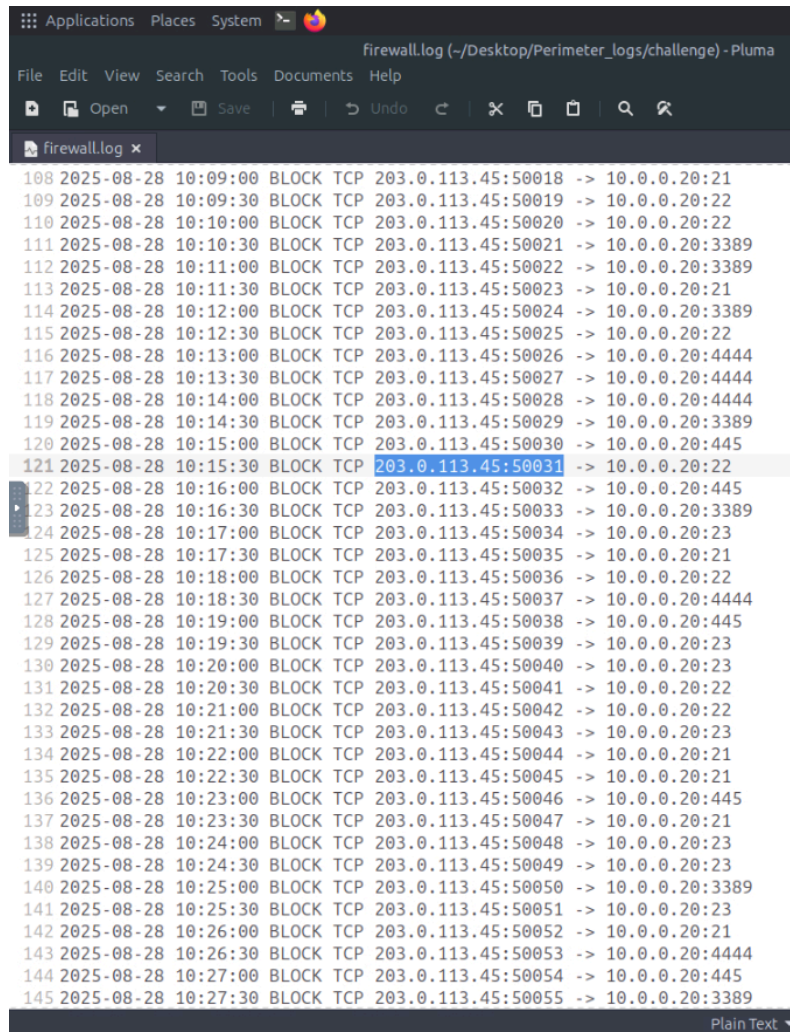
Figure 5: VPN log — brute-force authentication attempts from 45.137.22.13

```
44 2025-09-22 10:12:44 FAILED_AUTH TCP 45.137.22.13:31275 -> 10.0.0.1:443 (user 'test1')
45 2025-09-22 10:12:45 FAILED_AUTH TCP 45.137.22.13:31274 -> 10.0.0.1:443 (user 'user1')
46 2025-09-22 10:12:46 SUCCESS_AUTH TCP 198.51.100.88:41238 -> 10.0.0.1:443 (user 'b.jones')
47 2025-09-22 10:12:47 FAILED_AUTH TCP 45.137.22.13:31275 -> 10.0.0.1:443 (user 'test2')
48 2025-09-22 10:12:48 FAILED_AUTH TCP 45.137.22.13:31276 -> 10.0.0.1:443 (user 'user2')
49 2025-09-22 10:12:49 SUCCESS_AUTH TCP 203.0.113.55:51239 -> 10.0.0.1:443 (user 's.smith')
50 2025-09-22 10:12:50 FAILED_AUTH TCP 45.137.22.13:31277 -> 10.0.0.1:443 (user 'guest1')
51 2025-09-22 10:12:51 FAILED_AUTH TCP 45.137.22.13:31278 -> 10.0.0.1:443 (user 'ftpuser')
52 2025-09-22 10:12:52 SUCCESS_AUTH TCP 192.0.2.140:50004 -> 10.0.0.1:443 (user 'm.doe')
53 2025-09-22 10:12:53 FAILED_AUTH TCP 45.137.22.13:31279 -> 10.0.0.1:443 (user 'webmaster')
54 2025-09-22 10:12:54 FAILED_AUTH TCP 45.137.22.13:31280 -> 10.0.0.1:443 (user 'sales')
55 2025-09-22 10:12:55 SUCCESS_AUTH TCP 198.51.100.88:41239 -> 10.0.0.1:443 (user 'j.carter')
56 2025-09-22 10:12:56 FAILED_AUTH TCP 45.137.22.13:31281 -> 10.0.0.1:443 (user 'info')
57 2025-09-22 10:12:57 FAILED_AUTH TCP 45.137.22.13:31282 -> 10.0.0.1:443 (user 'demo')
58 2025-09-22 10:12:58 SUCCESS_AUTH TCP 203.0.113.55:51240 -> 10.0.0.1:443 (user 'a.williams')
59 2025-09-22 10:12:59 FAILED_AUTH TCP 45.137.22.13:31283 -> 10.0.0.1:443 (user 'login')
60 2025-09-22 10:13:00 FAILED_AUTH TCP 45.137.22.13:31284 -> 10.0.0.1:443 (user 'student')
61 2025-09-22 10:13:01 SUCCESS_AUTH TCP 192.0.2.140:50005 -> 10.0.0.1:443 (user 'b.jones')
62 2025-09-22 10:13:02 FAILED_AUTH TCP 45.137.22.13:31285 -> 10.0.0.1:443 (user 'manager')
63 2025-09-22 10:13:03 FAILED_AUTH TCP 45.137.22.13:31286 -> 10.0.0.1:443 (user 'staff')
64 2025-09-22 10:13:04 SUCCESS_AUTH TCP 198.51.100.88:41240 -> 10.0.0.1:443 (user 's.smith')
65 2025-09-22 10:13:05 FAILED_AUTH TCP 45.137.22.13:31287 -> 10.0.0.1:443 (user 'hr')
66 2025-09-22 10:13:06 FAILED_AUTH TCP 45.137.22.13:31288 -> 10.0.0.1:443 (user 'it')
67 2025-09-22 10:13:07 SUCCESS_AUTH TCP 203.0.113.55:51241 -> 10.0.0.1:443 (user 'm.doe')
68 2025-09-22 10:13:08 FAILED_AUTH TCP 45.137.22.13:31289 -> 10.0.0.1:443 (user 'marketing')
69 2025-09-22 10:13:09 FAILED_AUTH TCP 45.137.22.13:31290 -> 10.0.0.1:443 (user 'finance')
70 2025-09-22 10:13:10 SUCCESS_AUTH TCP 192.0.2.140:50006 -> 10.0.0.1:443 (user 'j.carter')
71 2025-09-22 10:13:11 FAILED_AUTH TCP 45.137.22.13:31291 -> 10.0.0.1:443 (user 'billing')
72 2025-09-22 10:13:12 FAILED_AUTH TCP 45.137.22.13:31292 -> 10.0.0.1:443 (user 'john')
73 2025-09-22 10:13:13 SUCCESS_AUTH TCP 198.51.100.88:41241 -> 10.0.0.1:443 (user 'a.williams')
74 2025-09-22 10:13:14 FAILED_AUTH TCP 45.137.22.13:31293 -> 10.0.0.1:443 (user 'jane')
75 2025-09-22 10:13:15 FAILED_AUTH TCP 45.137.22.13:31294 -> 10.0.0.1:443 (user 'susan')
76 2025-09-22 10:13:16 SUCCESS_AUTH TCP 203.0.113.55:51242 -> 10.0.0.1:443 (user 'b.jones')
77 2025-09-22 10:13:17 FAILED_AUTH TCP 45.137.22.13:31295 -> 10.0.0.1:443 (user 'bob')
78 2025-09-22 10:13:18 FAILED_AUTH TCP 45.137.22.13:31296 -> 10.0.0.1:443 (user 'sally')
79 2025-09-22 10:13:19 SUCCESS_AUTH TCP 192.0.2.140:50007 -> 10.0.0.1:443 (user 's.smith')
80 2025-09-22 10:13:20 FAILED_AUTH TCP 45.137.22.13:31297 -> 10.0.0.1:443 (user 'temp')
81 2025-09-22 10:13:21 FAILED_AUTH TCP 45.137.22.13:31298 -> 10.0.0.1:443 (user 'scanner')
82 2025-09-22 10:13:22 SUCCESS_AUTH TCP 198.51.100.88:41242 -> 10.0.0.1:443 (user 'm.doe')
```

Figure 6: VPN log — continued credential stuffing, legitimate users authenticating from trusted IPs

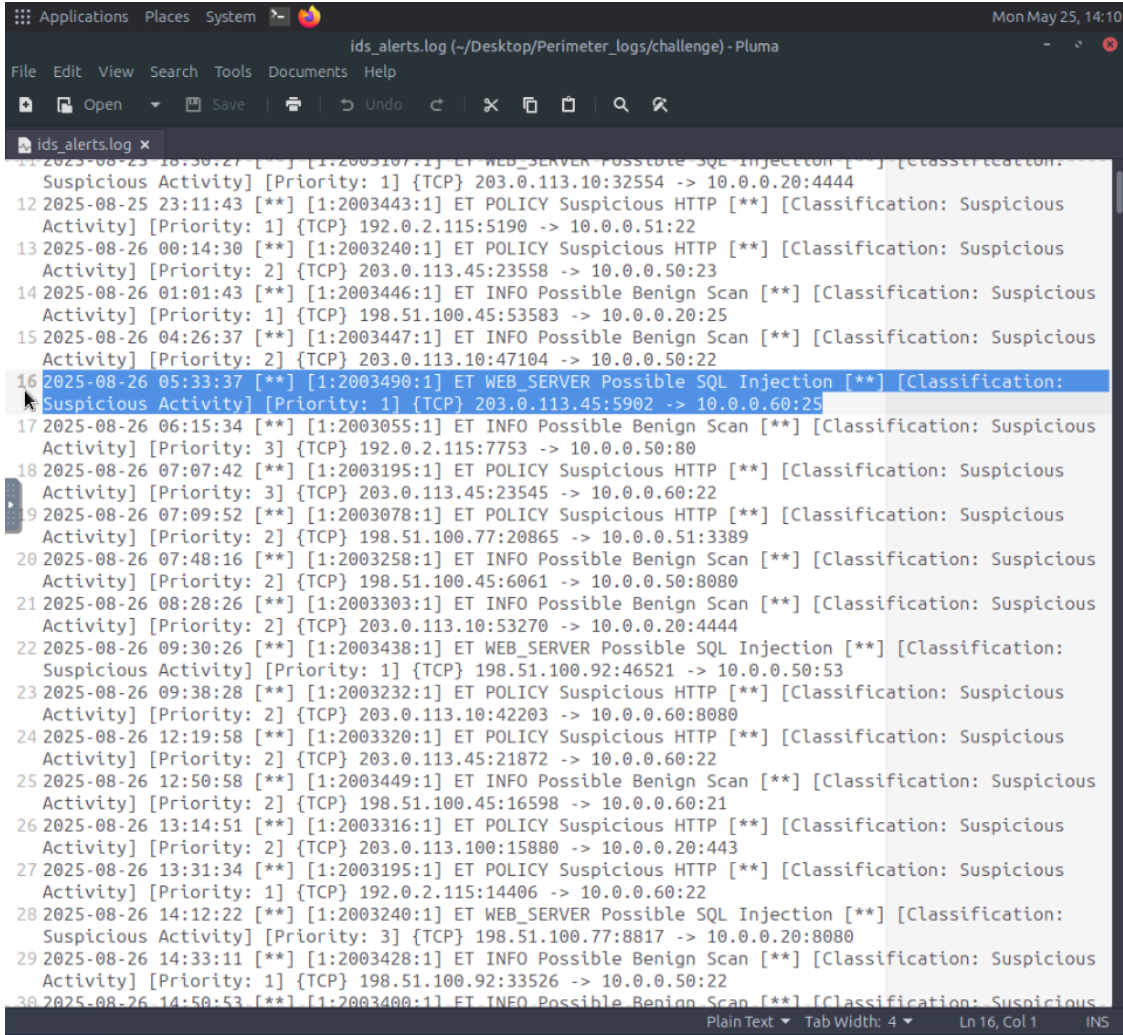
4.2 Legitimate vs Malicious Authentication

- Legitimate successful authentications originated from trusted IP ranges (192.0.2.x, 198.51.100.88, 203.0.113.55) for users s.smith, m.doe, j.carter, a.williams, and b.jones.
- All failed authentications came exclusively from 45.137.22.13, making it straightforward to identify and block the attacking IP.



```
firewall.log (~/Desktop/Perimeter_logs/challenge) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo X Copy Paste Find
firewall.log x
108 2025-08-28 10:09:00 BLOCK TCP 203.0.113.45:50018 -> 10.0.0.20:21
109 2025-08-28 10:09:30 BLOCK TCP 203.0.113.45:50019 -> 10.0.0.20:22
110 2025-08-28 10:10:00 BLOCK TCP 203.0.113.45:50020 -> 10.0.0.20:22
111 2025-08-28 10:10:30 BLOCK TCP 203.0.113.45:50021 -> 10.0.0.20:3389
112 2025-08-28 10:11:00 BLOCK TCP 203.0.113.45:50022 -> 10.0.0.20:3389
113 2025-08-28 10:11:30 BLOCK TCP 203.0.113.45:50023 -> 10.0.0.20:21
114 2025-08-28 10:12:00 BLOCK TCP 203.0.113.45:50024 -> 10.0.0.20:3389
115 2025-08-28 10:12:30 BLOCK TCP 203.0.113.45:50025 -> 10.0.0.20:22
116 2025-08-28 10:13:00 BLOCK TCP 203.0.113.45:50026 -> 10.0.0.20:4444
117 2025-08-28 10:13:30 BLOCK TCP 203.0.113.45:50027 -> 10.0.0.20:4444
118 2025-08-28 10:14:00 BLOCK TCP 203.0.113.45:50028 -> 10.0.0.20:4444
119 2025-08-28 10:14:30 BLOCK TCP 203.0.113.45:50029 -> 10.0.0.20:3389
120 2025-08-28 10:15:00 BLOCK TCP 203.0.113.45:50030 -> 10.0.0.20:445
121 2025-08-28 10:15:30 BLOCK TCP 203.0.113.45:50031 -> 10.0.0.20:22
122 2025-08-28 10:16:00 BLOCK TCP 203.0.113.45:50032 -> 10.0.0.20:445
123 2025-08-28 10:16:30 BLOCK TCP 203.0.113.45:50033 -> 10.0.0.20:3389
124 2025-08-28 10:17:00 BLOCK TCP 203.0.113.45:50034 -> 10.0.0.20:23
125 2025-08-28 10:17:30 BLOCK TCP 203.0.113.45:50035 -> 10.0.0.20:21
126 2025-08-28 10:18:00 BLOCK TCP 203.0.113.45:50036 -> 10.0.0.20:22
127 2025-08-28 10:18:30 BLOCK TCP 203.0.113.45:50037 -> 10.0.0.20:4444
128 2025-08-28 10:19:00 BLOCK TCP 203.0.113.45:50038 -> 10.0.0.20:445
129 2025-08-28 10:19:30 BLOCK TCP 203.0.113.45:50039 -> 10.0.0.20:23
130 2025-08-28 10:20:00 BLOCK TCP 203.0.113.45:50040 -> 10.0.0.20:23
131 2025-08-28 10:20:30 BLOCK TCP 203.0.113.45:50041 -> 10.0.0.20:22
132 2025-08-28 10:21:00 BLOCK TCP 203.0.113.45:50042 -> 10.0.0.20:22
133 2025-08-28 10:21:30 BLOCK TCP 203.0.113.45:50043 -> 10.0.0.20:23
134 2025-08-28 10:22:00 BLOCK TCP 203.0.113.45:50044 -> 10.0.0.20:21
135 2025-08-28 10:22:30 BLOCK TCP 203.0.113.45:50045 -> 10.0.0.20:21
136 2025-08-28 10:23:00 BLOCK TCP 203.0.113.45:50046 -> 10.0.0.20:445
137 2025-08-28 10:23:30 BLOCK TCP 203.0.113.45:50047 -> 10.0.0.20:21
138 2025-08-28 10:24:00 BLOCK TCP 203.0.113.45:50048 -> 10.0.0.20:23
139 2025-08-28 10:24:30 BLOCK TCP 203.0.113.45:50049 -> 10.0.0.20:23
140 2025-08-28 10:25:00 BLOCK TCP 203.0.113.45:50050 -> 10.0.0.20:3389
141 2025-08-28 10:25:30 BLOCK TCP 203.0.113.45:50051 -> 10.0.0.20:23
142 2025-08-28 10:26:00 BLOCK TCP 203.0.113.45:50052 -> 10.0.0.20:21
143 2025-08-28 10:26:30 BLOCK TCP 203.0.113.45:50053 -> 10.0.0.20:4444
144 2025-08-28 10:27:00 BLOCK TCP 203.0.113.45:50054 -> 10.0.0.20:445
145 2025-08-28 10:27:30 BLOCK TCP 203.0.113.45:50055 -> 10.0.0.20:3389
Plain Text
```

Figure 7: VPN log — pattern distinguishing legitimate users from brute-force attacker



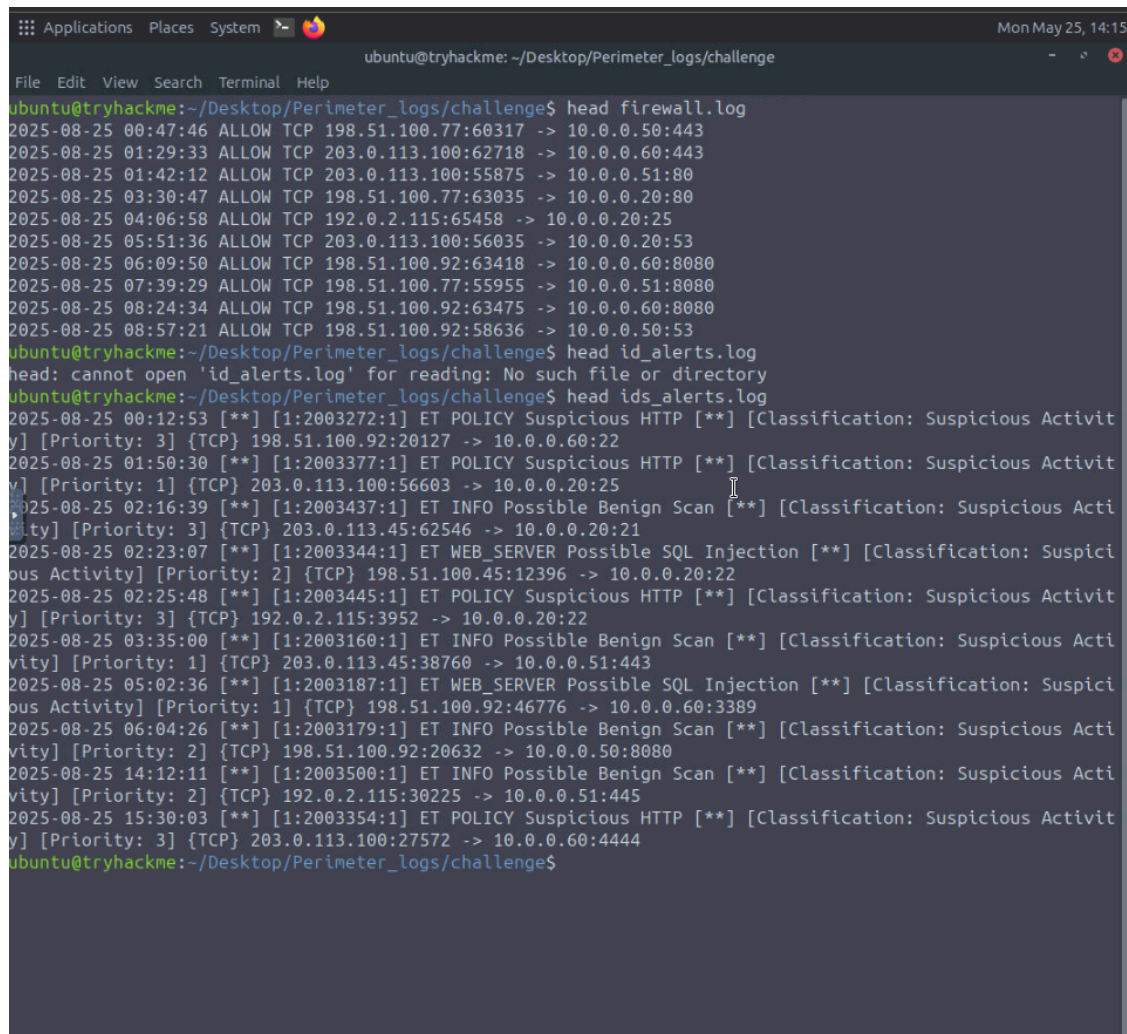
```
ids_alerts.log (-/Desktop/Perimeter_logs/challenge) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo Ctrl+Z Ctrl+Y Ctrl+F
ids_alerts.log x
11 2025-08-25 18:30:27 [**] [1:2003107:1] ET WEB_SERVER Possible SQL Injection [**] [Classification:
Suspicious Activity] [Priority: 1] {TCP} 203.0.113.10:32554 -> 10.0.0.20:4444
12 2025-08-25 23:11:43 [**] [1:2003443:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 1] {TCP} 192.0.2.115:5190 -> 10.0.0.51:22
13 2025-08-26 00:14:30 [**] [1:2003240:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.45:23558 -> 10.0.0.50:23
14 2025-08-26 01:01:43 [**] [1:2003446:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 1] {TCP} 198.51.100.45:53583 -> 10.0.0.20:25
15 2025-08-26 04:26:37 [**] [1:2003447:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.10:47104 -> 10.0.0.50:22
16 2025-08-26 05:33:37 [**] [1:2003490:1] ET WEB_SERVER Possible SQL Injection [**] [Classification:
Suspicious Activity] [Priority: 1] {TCP} 203.0.113.45:5902 -> 10.0.0.60:25
17 2025-08-26 06:15:34 [**] [1:2003055:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 3] {TCP} 192.0.2.115:7753 -> 10.0.0.50:80
18 2025-08-26 07:07:42 [**] [1:2003195:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 3] {TCP} 203.0.113.45:23545 -> 10.0.0.60:22
19 2025-08-26 07:09:52 [**] [1:2003078:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 198.51.100.77:20865 -> 10.0.0.51:3389
20 2025-08-26 07:48:16 [**] [1:2003258:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 198.51.100.45:6061 -> 10.0.0.50:8080
21 2025-08-26 08:28:26 [**] [1:2003303:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.10:53270 -> 10.0.0.20:4444
22 2025-08-26 09:30:26 [**] [1:2003438:1] ET WEB_SERVER Possible SQL Injection [**] [Classification:
Suspicious Activity] [Priority: 1] {TCP} 198.51.100.92:46521 -> 10.0.0.50:53
23 2025-08-26 09:38:28 [**] [1:2003232:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.10:42203 -> 10.0.0.60:8080
24 2025-08-26 12:19:58 [**] [1:2003320:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.45:21872 -> 10.0.0.60:22
25 2025-08-26 12:50:58 [**] [1:2003449:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 198.51.100.45:16598 -> 10.0.0.60:21
26 2025-08-26 13:14:51 [**] [1:2003316:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 2] {TCP} 203.0.113.100:15880 -> 10.0.0.20:443
27 2025-08-26 13:31:34 [**] [1:2003195:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious
Activity] [Priority: 1] {TCP} 192.0.2.115:14406 -> 10.0.0.60:22
28 2025-08-26 14:12:22 [**] [1:2003240:1] ET WEB_SERVER Possible SQL Injection [**] [Classification:
Suspicious Activity] [Priority: 3] {TCP} 198.51.100.77:8817 -> 10.0.0.20:8080
29 2025-08-26 14:33:11 [**] [1:2003428:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 1] {TCP} 198.51.100.92:33526 -> 10.0.0.50:22
30 2025-08-26 14:50:53 [**] [1:2003400:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious
Activity] [Priority: 1] {TCP} 198.51.100.92:33526 -> 10.0.0.50:22
Plain Text Tab Width: 4 Ln 16, Col 1 INS
```

Figure 8: VPN log — additional failed auth attempts showing username enumeration

5. Firewall and IDS Log Analysis

5.1 Firewall Log — Terminal Analysis

- The challenge directory was explored using the terminal to analyse firewall.log and ids_alerts.log.
- The head command was used to preview both log files.
- The firewall log showed standard ALLOW traffic patterns.
- While the IDS alert log contained Suricata-format alerts including ET POLICY Suspicious HTTP, ET INFO Possible Benign Scan, ET WEB_SERVER Possible SQL Injection, and ET POLICY Suspicious HTTP classifications.

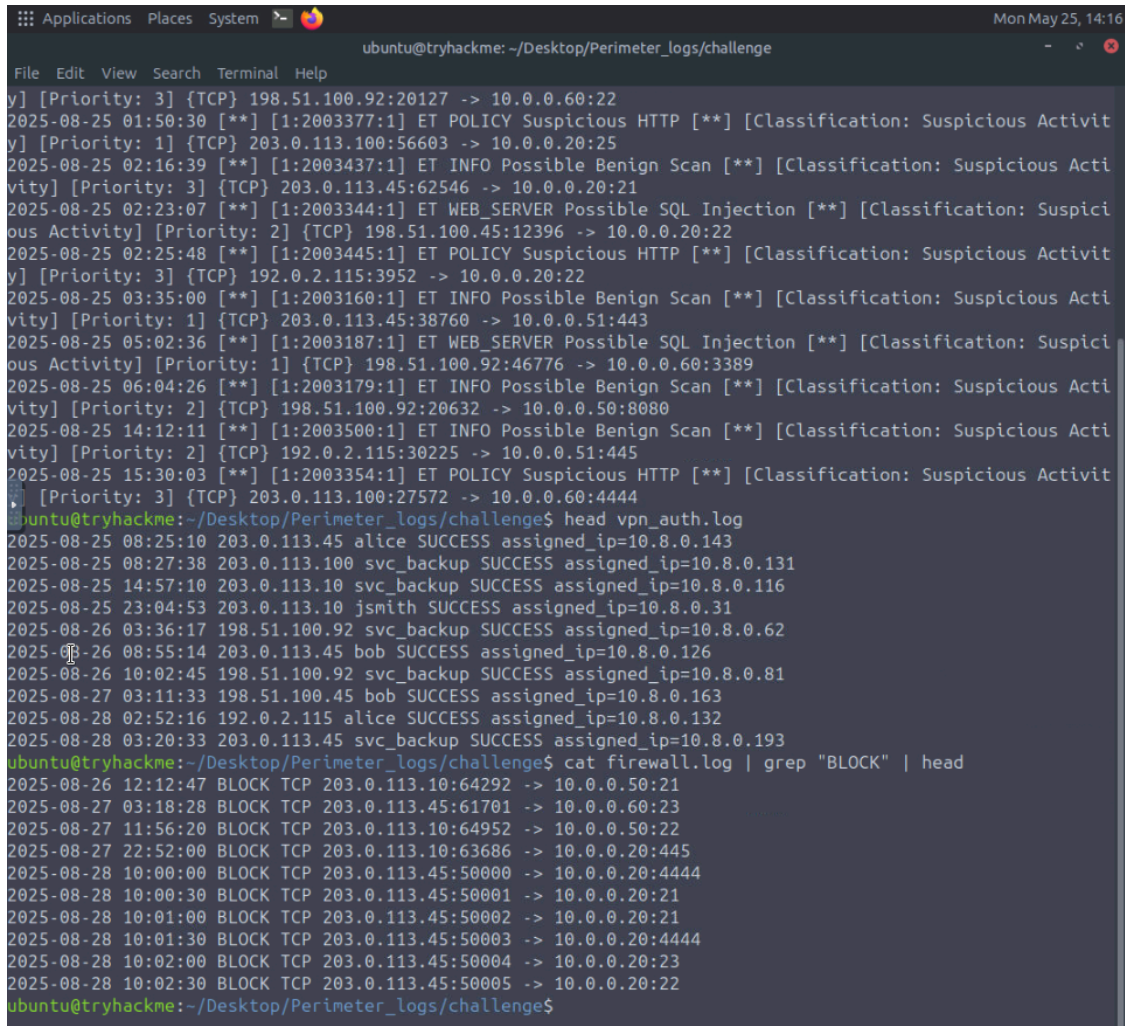


```
ubuntu@tryhackme: ~/Desktop/Perimeter_logs/challenge
File Edit View Search Terminal Help
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head firewall.log
2025-08-25 00:47:46 ALLOW TCP 198.51.100.77:60317 -> 10.0.0.50:443
2025-08-25 01:29:33 ALLOW TCP 203.0.113.100:62718 -> 10.0.0.60:443
2025-08-25 01:42:12 ALLOW TCP 203.0.113.100:55875 -> 10.0.0.51:80
2025-08-25 03:30:47 ALLOW TCP 198.51.100.77:63035 -> 10.0.0.20:80
2025-08-25 04:06:58 ALLOW TCP 192.0.2.115:65458 -> 10.0.0.20:25
2025-08-25 05:51:36 ALLOW TCP 203.0.113.100:56035 -> 10.0.0.20:53
2025-08-25 06:09:50 ALLOW TCP 198.51.100.92:63418 -> 10.0.0.60:8080
2025-08-25 07:39:29 ALLOW TCP 198.51.100.77:55955 -> 10.0.0.51:8080
2025-08-25 08:24:34 ALLOW TCP 198.51.100.92:63475 -> 10.0.0.60:8080
2025-08-25 08:57:21 ALLOW TCP 198.51.100.92:58636 -> 10.0.0.50:53
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head id_alerts.log
head: cannot open 'id_alerts.log' for reading: No such file or directory
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head ids_alerts.log
2025-08-25 00:12:53 [**] [1:2003272:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 198.51.100.92:20127 -> 10.0.0.60:22
2025-08-25 01:50:30 [**] [1:2003377:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 203.0.113.100:56603 -> 10.0.0.20:25
2025-08-25 02:16:39 [**] [1:2003437:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 203.0.113.45:62546 -> 10.0.0.20:21
2025-08-25 02:23:07 [**] [1:2003344:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 198.51.100.45:12396 -> 10.0.0.20:22
2025-08-25 02:25:48 [**] [1:2003445:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 192.0.2.115:3952 -> 10.0.0.20:22
2025-08-25 03:35:00 [**] [1:2003160:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 203.0.113.45:38760 -> 10.0.0.51:443
2025-08-25 05:02:36 [**] [1:2003187:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 198.51.100.92:46776 -> 10.0.0.60:3389
2025-08-25 06:04:26 [**] [1:2003179:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 198.51.100.92:20632 -> 10.0.0.50:8080
2025-08-25 14:12:11 [**] [1:2003500:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 192.0.2.115:30225 -> 10.0.0.51:445
2025-08-25 15:30:03 [**] [1:2003354:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 203.0.113.100:27572 -> 10.0.0.60:4444
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 10: Terminal — head firewall.log and head ids_alerts.log showing IDS alert format

5.2 Firewall Log — Suspicious IP Investigation

- The firewall log was filtered to identify traffic patterns from specific IP addresses.
- Analysis of traffic from 203.0.113.100 revealed consistent ALLOW entries to internal hosts on ports 443, 80, 3389, and 8080
- This is indicating a potential internal reconnaissance or lateral movement pattern across multiple services.



```
ubuntu@tryhackme: ~/Desktop/Perimeter_logs/challenge
y] [Priority: 3] {TCP} 198.51.100.92:20127 -> 10.0.0.60:22
2025-08-25 01:50:30 [**] [1:2003377:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 203.0.113.100:56603 -> 10.0.0.20:25
2025-08-25 02:16:39 [**] [1:2003437:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 203.0.113.45:62546 -> 10.0.0.20:21
2025-08-25 02:23:07 [**] [1:2003344:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 198.51.100.45:12396 -> 10.0.0.20:22
2025-08-25 02:25:48 [**] [1:2003445:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 192.0.2.115:3952 -> 10.0.0.20:22
2025-08-25 03:35:00 [**] [1:2003160:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 203.0.113.45:38760 -> 10.0.0.51:443
2025-08-25 05:02:36 [**] [1:2003187:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 198.51.100.92:46776 -> 10.0.0.60:3389
2025-08-25 06:04:26 [**] [1:2003179:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 198.51.100.92:20632 -> 10.0.0.50:8080
2025-08-25 14:12:11 [**] [1:2003500:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 2] {TCP} 192.0.2.115:30225 -> 10.0.0.51:445
2025-08-25 15:30:03 [**] [1:2003354:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 203.0.113.100:27572 -> 10.0.0.60:4444
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head vpn_auth.log
2025-08-25 08:25:10 203.0.113.45 alice SUCCESS assigned_ip=10.8.0.143
2025-08-25 08:27:38 203.0.113.100 svc_backup SUCCESS assigned_ip=10.8.0.131
2025-08-25 14:57:10 203.0.113.10 svc_backup SUCCESS assigned_ip=10.8.0.116
2025-08-25 23:04:53 203.0.113.10 jsmith SUCCESS assigned_ip=10.8.0.31
2025-08-26 03:36:17 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.62
2025-08-26 08:55:14 203.0.113.45 bob SUCCESS assigned_ip=10.8.0.126
2025-08-26 10:02:45 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.81
2025-08-27 03:11:33 198.51.100.45 bob SUCCESS assigned_ip=10.8.0.163
2025-08-28 02:52:16 192.0.2.115 alice SUCCESS assigned_ip=10.8.0.132
2025-08-28 03:20:33 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.193
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP 203.0.113.45:61701 -> 10.0.0.60:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
2025-08-27 22:52:00 BLOCK TCP 203.0.113.10:63686 -> 10.0.0.20:445
2025-08-28 10:00:00 BLOCK TCP 203.0.113.45:50000 -> 10.0.0.20:4444
2025-08-28 10:00:30 BLOCK TCP 203.0.113.45:50001 -> 10.0.0.20:21
2025-08-28 10:01:00 BLOCK TCP 203.0.113.45:50002 -> 10.0.0.20:21
2025-08-28 10:01:30 BLOCK TCP 203.0.113.45:50003 -> 10.0.0.20:4444
2025-08-28 10:02:00 BLOCK TCP 203.0.113.45:50004 -> 10.0.0.20:23
2025-08-28 10:02:30 BLOCK TCP 203.0.113.45:50005 -> 10.0.0.20:22
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 11: Firewall log — filtered traffic from 203.0.113.100 across multiple destination ports

```
Applications Places System ubuntu@tryhackme: ~/Desktop/Perimeter_logs/challenge Mon May 25, 14:18
File Edit View Search Terminal Help
2025-08-25 02:23:07 [**] [1:2003344:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspici
ous Activity] [Priority: 2] {TCP} 198.51.100.45:12396 -> 10.0.0.20:22
2025-08-25 02:25:48 [**] [1:2003445:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activit
y] [Priority: 3] {TCP} 192.0.2.115:3952 -> 10.0.0.20:22
2025-08-25 03:35:00 [**] [1:2003160:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Acti
vity] [Priority: 1] {TCP} 203.0.113.45:38760 -> 10.0.0.51:443
2025-08-25 05:02:36 [**] [1:2003187:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspici
ous Activity] [Priority: 1] {TCP} 198.51.100.92:46776 -> 10.0.0.60:3389
2025-08-25 06:04:26 [**] [1:2003179:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Acti
vity] [Priority: 2] {TCP} 198.51.100.92:20632 -> 10.0.0.50:8080
2025-08-25 14:12:11 [**] [1:2003500:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Acti
vity] [Priority: 2] {TCP} 192.0.2.115:30225 -> 10.0.0.51:445
2025-08-25 15:30:03 [**] [1:2003354:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activit
y] [Priority: 3] {TCP} 203.0.113.100:27572 -> 10.0.0.60:4444
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head vpn_auth.log
2025-08-25 08:25:10 203.0.113.45 alice SUCCESS assigned_ip=10.8.0.143
2025-08-25 08:27:38 203.0.113.100 svc_backup SUCCESS assigned_ip=10.8.0.131
2025-08-25 14:57:10 203.0.113.10 svc_backup SUCCESS assigned_ip=10.8.0.116
2025-08-25 23:04:53 203.0.113.10 jsmith SUCCESS assigned_ip=10.8.0.31
2025-08-26 03:36:17 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.62
2025-08-26 08:55:14 203.0.113.45 bob SUCCESS assigned_ip=10.8.0.126
2025-08-26 10:02:45 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.81
2025-08-27 03:11:33 198.51.100.45 bob SUCCESS assigned_ip=10.8.0.163
2025-08-28 02:52:16 192.0.2.115 alice SUCCESS assigned_ip=10.8.0.132
2025-08-28 03:20:33 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.193
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP 203.0.113.45:61701 -> 10.0.0.60:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
2025-08-27 22:52:00 BLOCK TCP 203.0.113.10:63686 -> 10.0.0.20:445
2025-08-28 10:00:00 BLOCK TCP 203.0.113.45:50000 -> 10.0.0.20:4444
2025-08-28 10:00:30 BLOCK TCP 203.0.113.45:50001 -> 10.0.0.20:21
2025-08-28 10:01:00 BLOCK TCP 203.0.113.45:50002 -> 10.0.0.20:21
2025-08-28 10:01:30 BLOCK TCP 203.0.113.45:50003 -> 10.0.0.20:4444
2025-08-28 10:02:00 BLOCK TCP 203.0.113.45:50004 -> 10.0.0.20:23
2025-08-28 10:02:30 BLOCK TCP 203.0.113.45:50005 -> 10.0.0.20:22
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | cut -d' ' -f5 | cu
t -d: -f1 | sort -nr | uniq -c
  279 203.0.113.45
   46 203.0.113.10
   26 10.8.0.23
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

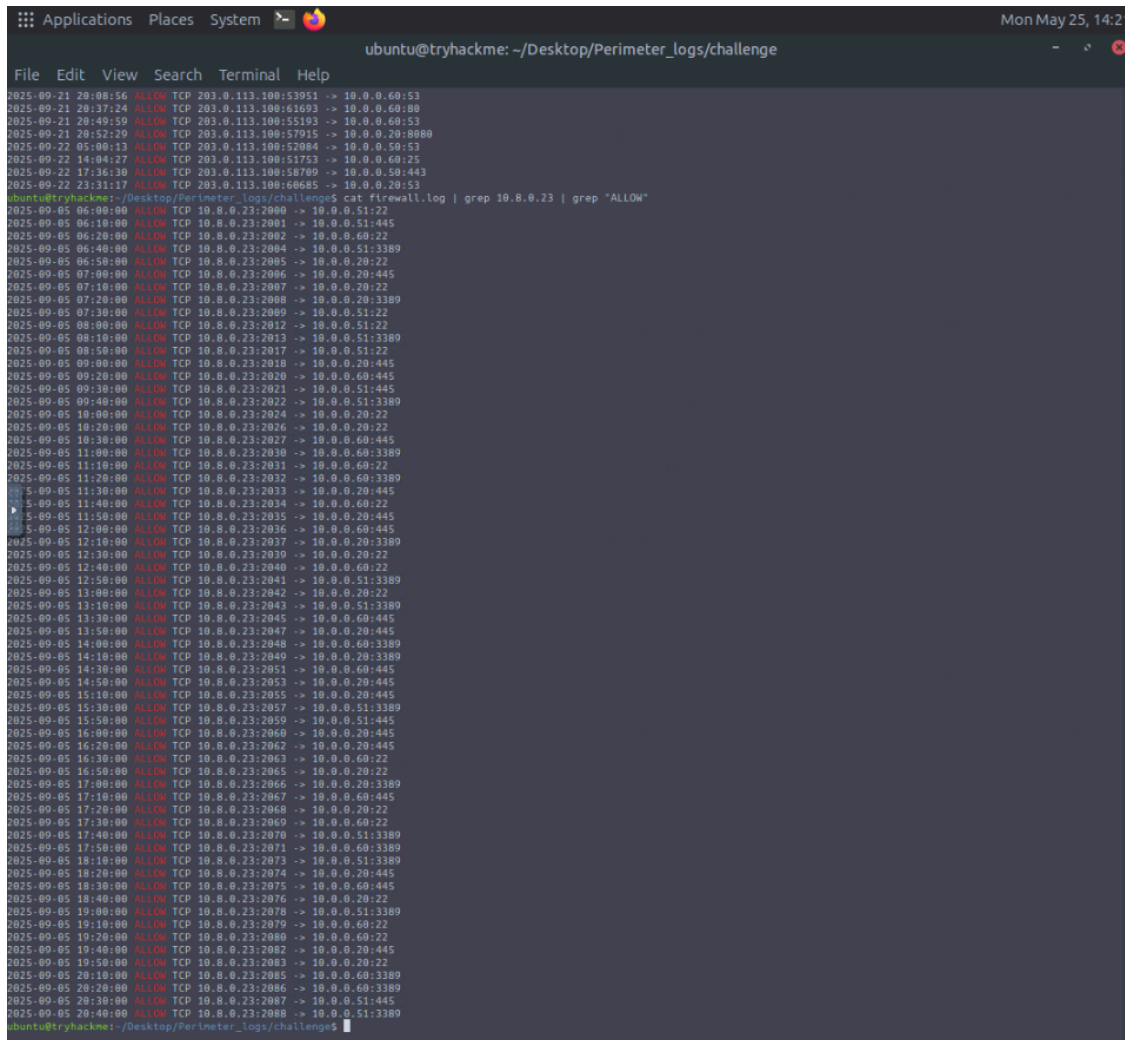
Figure 12: Firewall log — continued traffic analysis showing destination port variety


```
Applications Places System ubuntu@tryhackme: ~/Desktop/Perimeter_logs/challenge Mon May 25, 14:19
File Edit View Search Terminal Help
2025-08-26 03:36:17 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.62
2025-08-26 08:55:14 203.0.113.45 bob SUCCESS assigned_ip=10.8.0.126
2025-08-26 10:02:45 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.81
2025-08-27 03:11:33 198.51.100.45 bob SUCCESS assigned_ip=10.8.0.163
2025-08-28 02:52:16 192.0.2.115 alice SUCCESS assigned_ip=10.8.0.132
2025-08-28 03:20:33 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.193
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP 203.0.113.45:61701 -> 10.0.0.60:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
2025-08-27 22:52:00 BLOCK TCP 203.0.113.10:63686 -> 10.0.0.20:445
2025-08-28 10:00:00 BLOCK TCP 203.0.113.45:50000 -> 10.0.0.20:4444
2025-08-28 10:00:30 BLOCK TCP 203.0.113.45:50001 -> 10.0.0.20:21
2025-08-28 10:01:00 BLOCK TCP 203.0.113.45:50002 -> 10.0.0.20:21
2025-08-28 10:01:30 BLOCK TCP 203.0.113.45:50003 -> 10.0.0.20:4444
2025-08-28 10:02:00 BLOCK TCP 203.0.113.45:50004 -> 10.0.0.20:23
2025-08-28 10:02:30 BLOCK TCP 203.0.113.45:50005 -> 10.0.0.20:22
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | cut -d' ' -f5 | cu
-d: -f1 | sort -nr | uniq -c
279 203.0.113.45
46 203.0.113.10
26 10.8.0.23
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep 203.0.113.45 | grep "ALLOW"
2025-08-26 00:17:58 ALLOW TCP 203.0.113.45:61009 -> 10.0.0.60:4444
2025-08-26 22:04:34 ALLOW TCP 203.0.113.45:55996 -> 10.0.0.50:445
2025-08-27 21:04:23 ALLOW TCP 203.0.113.45:53944 -> 10.0.0.50:22
2025-08-28 15:50:50 ALLOW TCP 203.0.113.45:56123 -> 10.0.0.60:3389
2025-08-30 20:26:23 ALLOW TCP 203.0.113.45:61685 -> 10.0.0.20:4444
2025-09-02 09:25:06 ALLOW TCP 203.0.113.45:50550 -> 10.0.0.50:22
2025-09-03 02:19:40 ALLOW TCP 203.0.113.45:40118 -> 10.0.0.50:443
2025-09-03 02:19:50 ALLOW TCP 203.0.113.45:40119 -> 10.0.0.50:443
2025-09-06 02:49:07 ALLOW TCP 203.0.113.45:61832 -> 10.0.0.60:3389
2025-09-07 14:56:28 ALLOW TCP 203.0.113.45:51657 -> 10.0.0.50:23
2025-09-08 13:28:29 ALLOW TCP 203.0.113.45:55737 -> 10.0.0.50:21
2025-09-08 13:54:51 ALLOW TCP 203.0.113.45:58194 -> 10.0.0.20:4444
2025-09-10 18:17:46 ALLOW TCP 203.0.113.45:59715 -> 10.0.0.50:3389
2025-09-16 01:41:52 ALLOW TCP 203.0.113.45:58113 -> 10.0.0.51:21
2025-09-18 16:29:32 ALLOW TCP 203.0.113.45:52036 -> 10.0.0.60:3389
2025-09-21 15:00:31 ALLOW TCP 203.0.113.45:65151 -> 10.0.0.60:22
2025-09-22 15:46:02 ALLOW TCP 203.0.113.45:59771 -> 10.0.0.20:23
2025-09-22 17:22:11 ALLOW TCP 203.0.113.45:49360 -> 10.0.0.50:22
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 13: Firewall log — traffic correlation across source IPs

5.3 Firewall Log — ALLOW Traffic from Specific Host

- The grep command was used to filter ALLOW traffic from IP 10.8.0.23 to identify its communication patterns.
- The output showed high-frequency traffic across hourly intervals to multiple internal hosts on ports 22, 445, 3389, and 8080.
- This is consistent with an automated scanning tool performing internal network reconnaissance.



```
Applications Places System ubuntu@tryhackme: ~/Desktop/Perimeter_logs/challenge
File Edit View Search Terminal Help
2025-09-21 20:00:56 ALLOW TCP 203.0.113.100:53951 -> 10.0.0.60:53
2025-09-21 20:37:24 ALLOW TCP 203.0.113.100:61693 -> 10.0.0.60:80
2025-09-21 20:49:59 ALLOW TCP 203.0.113.100:55193 -> 10.0.0.60:53
2025-09-21 20:52:29 ALLOW TCP 203.0.113.100:57915 -> 10.0.0.20:8080
2025-09-22 05:00:13 ALLOW TCP 203.0.113.100:52084 -> 10.0.0.50:53
2025-09-22 14:04:27 ALLOW TCP 203.0.113.100:53753 -> 10.0.0.60:25
2025-09-22 17:36:30 ALLOW TCP 203.0.113.100:58769 -> 10.0.0.50:443
2025-09-22 23:31:17 ALLOW TCP 203.0.113.100:60685 -> 10.0.0.20:53
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep 10.8.0.23 | grep "ALLOW"
2025-09-05 06:00:00 ALLOW TCP 10.8.0.23:2080 -> 10.0.0.51:22
2025-09-05 06:10:00 ALLOW TCP 10.8.0.23:2081 -> 10.0.0.51:445
2025-09-05 06:20:00 ALLOW TCP 10.8.0.23:2082 -> 10.0.0.60:22
2025-09-05 06:40:00 ALLOW TCP 10.8.0.23:2084 -> 10.0.0.51:3389
2025-09-05 06:50:00 ALLOW TCP 10.8.0.23:2085 -> 10.0.0.20:22
2025-09-05 07:00:00 ALLOW TCP 10.8.0.23:2086 -> 10.0.0.20:445
2025-09-05 07:10:00 ALLOW TCP 10.8.0.23:2087 -> 10.0.0.20:22
2025-09-05 07:20:00 ALLOW TCP 10.8.0.23:2088 -> 10.0.0.20:3389
2025-09-05 07:30:00 ALLOW TCP 10.8.0.23:2089 -> 10.0.0.51:22
2025-09-05 08:00:00 ALLOW TCP 10.8.0.23:2012 -> 10.0.0.51:22
2025-09-05 08:10:00 ALLOW TCP 10.8.0.23:2013 -> 10.0.0.51:3389
2025-09-05 08:50:00 ALLOW TCP 10.8.0.23:2017 -> 10.0.0.51:22
2025-09-05 09:00:00 ALLOW TCP 10.8.0.23:2018 -> 10.0.0.20:445
2025-09-05 09:20:00 ALLOW TCP 10.8.0.23:2020 -> 10.0.0.60:445
2025-09-05 09:30:00 ALLOW TCP 10.8.0.23:2021 -> 10.0.0.51:445
2025-09-05 09:40:00 ALLOW TCP 10.8.0.23:2022 -> 10.0.0.51:3389
2025-09-05 10:00:00 ALLOW TCP 10.8.0.23:2024 -> 10.0.0.20:22
2025-09-05 10:20:00 ALLOW TCP 10.8.0.23:2026 -> 10.0.0.20:22
2025-09-05 10:30:00 ALLOW TCP 10.8.0.23:2027 -> 10.0.0.60:445
2025-09-05 11:00:00 ALLOW TCP 10.8.0.23:2030 -> 10.0.0.60:3389
2025-09-05 11:10:00 ALLOW TCP 10.8.0.23:2031 -> 10.0.0.60:22
2025-09-05 11:20:00 ALLOW TCP 10.8.0.23:2032 -> 10.0.0.60:3389
2025-09-05 11:30:00 ALLOW TCP 10.8.0.23:2033 -> 10.0.0.20:445
2025-09-05 11:40:00 ALLOW TCP 10.8.0.23:2034 -> 10.0.0.60:22
2025-09-05 11:50:00 ALLOW TCP 10.8.0.23:2035 -> 10.0.0.20:445
2025-09-05 12:00:00 ALLOW TCP 10.8.0.23:2036 -> 10.0.0.60:445
2025-09-05 12:10:00 ALLOW TCP 10.8.0.23:2037 -> 10.0.0.20:3389
2025-09-05 12:30:00 ALLOW TCP 10.8.0.23:2039 -> 10.0.0.20:22
2025-09-05 12:40:00 ALLOW TCP 10.8.0.23:2040 -> 10.0.0.60:22
2025-09-05 12:50:00 ALLOW TCP 10.8.0.23:2041 -> 10.0.0.51:3389
2025-09-05 13:00:00 ALLOW TCP 10.8.0.23:2042 -> 10.0.0.20:22
2025-09-05 13:10:00 ALLOW TCP 10.8.0.23:2043 -> 10.0.0.51:3389
2025-09-05 13:30:00 ALLOW TCP 10.8.0.23:2045 -> 10.0.0.60:445
2025-09-05 13:50:00 ALLOW TCP 10.8.0.23:2047 -> 10.0.0.20:445
2025-09-05 14:00:00 ALLOW TCP 10.8.0.23:2048 -> 10.0.0.60:3389
2025-09-05 14:10:00 ALLOW TCP 10.8.0.23:2049 -> 10.0.0.20:3389
2025-09-05 14:30:00 ALLOW TCP 10.8.0.23:2051 -> 10.0.0.60:445
2025-09-05 14:50:00 ALLOW TCP 10.8.0.23:2053 -> 10.0.0.20:445
2025-09-05 15:10:00 ALLOW TCP 10.8.0.23:2055 -> 10.0.0.20:445
2025-09-05 15:30:00 ALLOW TCP 10.8.0.23:2057 -> 10.0.0.51:3389
2025-09-05 15:50:00 ALLOW TCP 10.8.0.23:2059 -> 10.0.0.51:445
2025-09-05 16:00:00 ALLOW TCP 10.8.0.23:2060 -> 10.0.0.20:445
2025-09-05 16:20:00 ALLOW TCP 10.8.0.23:2062 -> 10.0.0.20:445
2025-09-05 16:30:00 ALLOW TCP 10.8.0.23:2063 -> 10.0.0.60:22
2025-09-05 16:50:00 ALLOW TCP 10.8.0.23:2065 -> 10.0.0.20:22
2025-09-05 17:00:00 ALLOW TCP 10.8.0.23:2066 -> 10.0.0.20:3389
2025-09-05 17:10:00 ALLOW TCP 10.8.0.23:2067 -> 10.0.0.60:445
2025-09-05 17:20:00 ALLOW TCP 10.8.0.23:2068 -> 10.0.0.20:22
2025-09-05 17:30:00 ALLOW TCP 10.8.0.23:2069 -> 10.0.0.60:22
2025-09-05 17:40:00 ALLOW TCP 10.8.0.23:2070 -> 10.0.0.51:3389
2025-09-05 17:50:00 ALLOW TCP 10.8.0.23:2071 -> 10.0.0.60:3389
2025-09-05 18:10:00 ALLOW TCP 10.8.0.23:2073 -> 10.0.0.51:3389
2025-09-05 18:20:00 ALLOW TCP 10.8.0.23:2074 -> 10.0.0.20:445
2025-09-05 18:30:00 ALLOW TCP 10.8.0.23:2075 -> 10.0.0.60:445
2025-09-05 18:40:00 ALLOW TCP 10.8.0.23:2076 -> 10.0.0.20:22
2025-09-05 19:00:00 ALLOW TCP 10.8.0.23:2078 -> 10.0.0.51:3389
2025-09-05 19:10:00 ALLOW TCP 10.8.0.23:2079 -> 10.0.0.60:22
2025-09-05 19:20:00 ALLOW TCP 10.8.0.23:2080 -> 10.0.0.60:22
2025-09-05 19:40:00 ALLOW TCP 10.8.0.23:2082 -> 10.0.0.20:445
2025-09-05 19:50:00 ALLOW TCP 10.8.0.23:2083 -> 10.0.0.20:22
2025-09-05 20:10:00 ALLOW TCP 10.8.0.23:2085 -> 10.0.0.60:3389
2025-09-05 20:20:00 ALLOW TCP 10.8.0.23:2086 -> 10.0.0.60:3389
2025-09-05 20:30:00 ALLOW TCP 10.8.0.23:2087 -> 10.0.0.51:445
2025-09-05 20:40:00 ALLOW TCP 10.8.0.23:2088 -> 10.0.0.51:3389
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 15: Firewall log — high-frequency ALLOW traffic from 10.8.0.23 indicating internal scan

5.4 IDS Alerts — C2 Beacons Detection

- IDS alert analysis identified ET TROJAN Possible C2 Beacons alerts originating from internal host 10.0.0.60 communicating to external IP 198.51.100.7 on port 4444
- This is a common C2 callback port.
- The beaconing occurred at regular 6-hour intervals (01:00, 07:00, 13:00, 19:00), a strong indicator of malware command-and-control activity.

```
2025-09-13 10:13:34 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.131
2025-09-13 12:08:26 203.0.113.45 bob SUCCESS assigned_ip=10.0.0.41
2025-09-15 09:29:17 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.52
2025-09-15 11:23:16 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.155
2025-09-17 13:17:32 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.78
2025-09-18 09:18:46 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.190
2025-09-19 05:33:15 203.0.113.45 bob SUCCESS assigned_ip=10.0.0.12
2025-09-21 06:59:25 203.0.113.45 bob SUCCESS assigned_ip=10.0.0.148
2025-09-22 08:09:24 203.0.113.45 alice SUCCESS assigned_ip=10.0.0.161
ubuntu@tryhackme:~/Desktop/Perimeter_Logs/challenge$ cat firewall.log | grep 203.0.113.45 | grep "ALLOW" | head
2025-08-26 00:17:58 ALLOW TCP 203.0.113.45:61009 -> 10.0.0.60:4444
2025-08-26 22:04:34 ALLOW TCP 203.0.113.45:55996 -> 10.0.0.50:445
2025-08-27 21:04:23 ALLOW TCP 203.0.113.45:53044 -> 10.0.0.50:22
2025-08-28 15:50:50 ALLOW TCP 203.0.113.45:56123 -> 10.0.0.60:3380
2025-08-30 20:26:23 ALLOW TCP 203.0.113.45:61605 -> 10.0.0.20:4444
2025-09-02 09:25:06 ALLOW TCP 203.0.113.45:50550 -> 10.0.0.50:22
2025-09-03 02:19:40 ALLOW TCP 203.0.113.45:40118 -> 10.0.0.50:443
2025-09-03 02:19:50 ALLOW TCP 203.0.113.45:40119 -> 10.0.0.50:443
2025-09-06 02:40:07 ALLOW TCP 203.0.113.45:61832 -> 10.0.0.60:3380
2025-09-07 14:56:28 ALLOW TCP 203.0.113.45:51657 -> 10.0.0.50:23
ubuntu@tryhackme:~/Desktop/Perimeter_Logs/challenge$ cat ids_alerts.log | grep 203.0.113.45 | grep "ALLOW" | head
2025-08-25 02:16:39 [**] [1:200337:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 3] [TCP] 203.0.113.45:62546 -> 10.0.0.20:21
2025-08-25 03:35:00 [**] [1:200316:0] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 1] [TCP] 203.0.113.45:38760 -> 10.0.0.51:443
2025-08-26 00:14:30 [**] [1:200324:0] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 2] [TCP] 203.0.113.45:23558 -> 10.0.0.50:23
2025-08-26 05:33:37 [**] [1:200349:0] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 1] [TCP] 203.0.113.45:5902 -> 10.0.0.60:25
2025-08-26 07:07:42 [**] [1:200319:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] [TCP] 203.0.113.45:23545 -> 10.0.0.60:22
2025-08-26 12:19:58 [**] [1:200329:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 2] [TCP] 203.0.113.45:21872 -> 10.0.0.60:22
2025-08-26 17:35:20 [**] [1:200325:0] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 1] [TCP] 203.0.113.45:26502 -> 10.0.0.50:443
2025-08-26 19:49:54 [**] [1:200313:1] ET WEB_SERVER Possible SQL Injection [**] [Classification: Suspicious Activity] [Priority: 1] [TCP] 203.0.113.45:63302 -> 10.0.0.60:443
2025-08-26 23:31:32 [**] [1:200323:1] ET INFO Possible Benign Scan [**] [Classification: Suspicious Activity] [Priority: 2] [TCP] 203.0.113.45:33332 -> 10.0.0.60:443
2025-08-27 02:25:35 [**] [1:200338:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 2] [TCP] 203.0.113.45:7348 -> 10.0.0.20:443
ubuntu@tryhackme:~/Desktop/Perimeter_Logs/challenge$ cat ids_alerts.log | grep -n 203.0.113.45 | grep 'SMB' | cut -d ' ' -f6,7,8,9,10,11,12 | head
2025-09-11 01:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30800 -> 198.51.100.7
7:4444
2025-09-11 07:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30801 -> 198.51.100.7
7:4444
2025-09-11 13:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30802 -> 198.51.100.7
7:4444
2025-09-11 19:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30803 -> 198.51.100.7
7:4444
2025-09-12 01:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30804 -> 198.51.100.7
7:4444
2025-09-12 07:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30805 -> 198.51.100.7
7:4444
2025-09-12 13:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30806 -> 198.51.100.7
7:4444
2025-09-12 19:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30807 -> 198.51.100.7
7:4444
2025-09-13 01:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30808 -> 198.51.100.7
7:4444
2025-09-13 07:00:00 [**] [1:200100:1] ET TROJAN Possible C2 Beacons [**] [Classification: A network Trojan was detected] [Priority: 1] [TCP] 10.0.0.60:30809 -> 198.51.100.7
7:4444
ubuntu@tryhackme:~/Desktop/Perimeter_Logs/challenge$
```

Figure 20: IDS alert — ET TROJAN C2 beaconing from 10.0.0.60 to 198.51.100.7:4444 at 6-hour intervals

5.5 Short-Answer Investigation Findings

- The following screenshots show terminal commands used to answer specific investigation questions in the lab, including filtering by IP, grep-based log parsing, and cross-referencing firewall with IDS logs:

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_alerts.log | grep -n 10.0.0.60 | grep 'SMB' | cut -d ' ' -f6,7,8,9,10,19,21 | head
14444
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2010 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2016 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2020 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2027 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2036 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2044 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2045 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2051 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2052 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2061 10.0.0.60:445
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 21: Terminal command — grep filter result

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_alerts.log | grep -n 198.51.100.77 | cut -d ' ' -f6,7,8,9,10,19,22,23 | head -n 15
2874x809
INFO Possible Benign Scan [**] 10.0.0.20:8080
INFO Possible Benign Scan [**] 10.0.0.20:4444
POLICY Suspicious HTTP [**] [Classification:
INFO Possible Benign Scan [**] 10.0.0.60:80
POLICY Suspicious HTTP [**] [Classification:
WEB_SERVER Possible SQL Injection [**] 10.0.0.51:21
INFO Possible Benign Scan [**] 10.0.0.60:4444
INFO Possible Benign Scan [**] 10.0.0.20:22
INFO Possible Benign Scan [**] 10.0.0.50:21
POLICY Suspicious HTTP [**] [Classification:
POLICY Suspicious HTTP [**] [Classification:
WEB_SERVER Possible SQL Injection [**] 10.0.0.50:21
INFO Possible Benign Scan [**] 10.0.0.20:53
INFO Possible Benign Scan [**] 10.0.0.50:21
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 22: Terminal command — cross-reference firewall and IDS logs

```
INFO Possible Benign Scan [**] 10.0.0.50:21
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_alerts.log | grep -n 198.51.100.77 | cut -d ' ' -f6,7,8,9,10,19,22,23 | uniq -c | sort -nr | head
8 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
7 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
3 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
3 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 23: Terminal — answer verification

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_alerts.log | grep -n 10.0.0.60 | cut -d ' ' -f6,7,8,9,10,19,22,23 | uniq -c | sort -nr | head
32 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
5 POLICY Suspicious HTTP [**] [Classification:
3 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
3 POLICY Suspicious HTTP [**] [Classification:
2 WEB_SERVER Possible SQL Injection [**] 10.0.0.60:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
2 TROJAN Possible C2 Beacons [**] (TCP) 198.51.100.77:4444
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

Figure 24: Terminal — additional log query result

6. Splunk SIEM Analysis

6.1 Ingesting Logs into Splunk

- All perimeter log sources were ingested into Splunk under the `network_logs` index.
- The Splunk search interface was used to query across `firewall_logs.json` and `ids_alerts.json` simultaneously.
- This enabled cross-source correlation that would not be possible analysing individual log files.

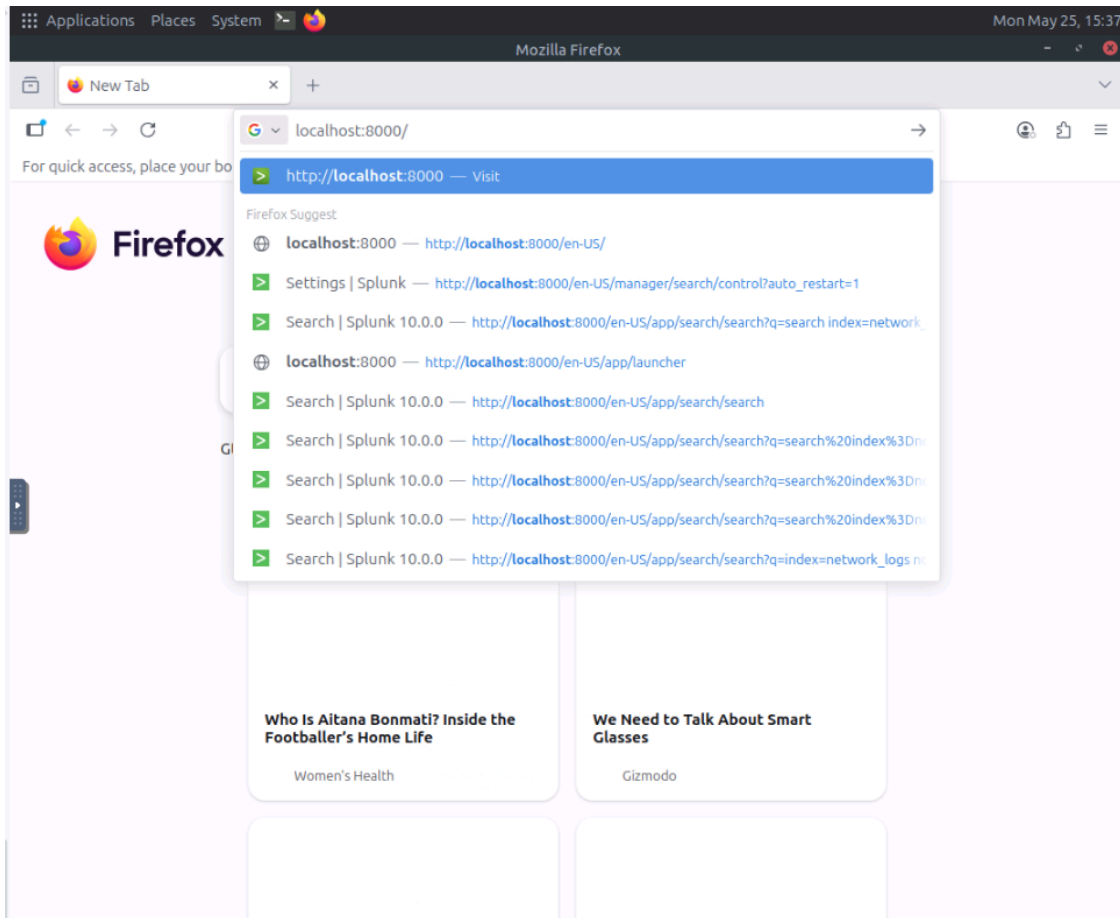


Figure 32: Splunk — search interface with `network_logs` index, firewall and IDS sources visible

6.2 IDS Alert Distribution

- A Splunk field analysis on the alert field revealed the top 10 alert types across the entire log dataset. The distribution showed:
- **ET INFO Possible Benign Scan**
 - 181 events (18.4%)
 - highest volume, indicating active scanning
- **ET WEB_SERVER Possible SQL Injection**
 - 167 events (17.0%)
 - sustained web application attacks
- **ET POLICY Suspicious HTTP**
 - 165 events (16.8%)
 - policy violations in HTTP traffic
- **ET POLICY VPN Authentication Failure**
 - 120 events (12.2%)
 - the brute-force VPN attack
- **ET SCAN Possible Portscan**
 - 120 events (12.2%)
 - network reconnaissance activity
- **ET TROJAN Possible C2 Beaconing**
 - 80 events (8.1%)
 - malware C2 communication
- **ET INFO Possible HTTP POST Large Upload**
 - 60 events (6.1%)
 - potential data exfiltration
- **ET EXPLOIT Possible MS-SMB Lateral Movement**
 - 32 events (3.3%)
 - lateral movement attempts

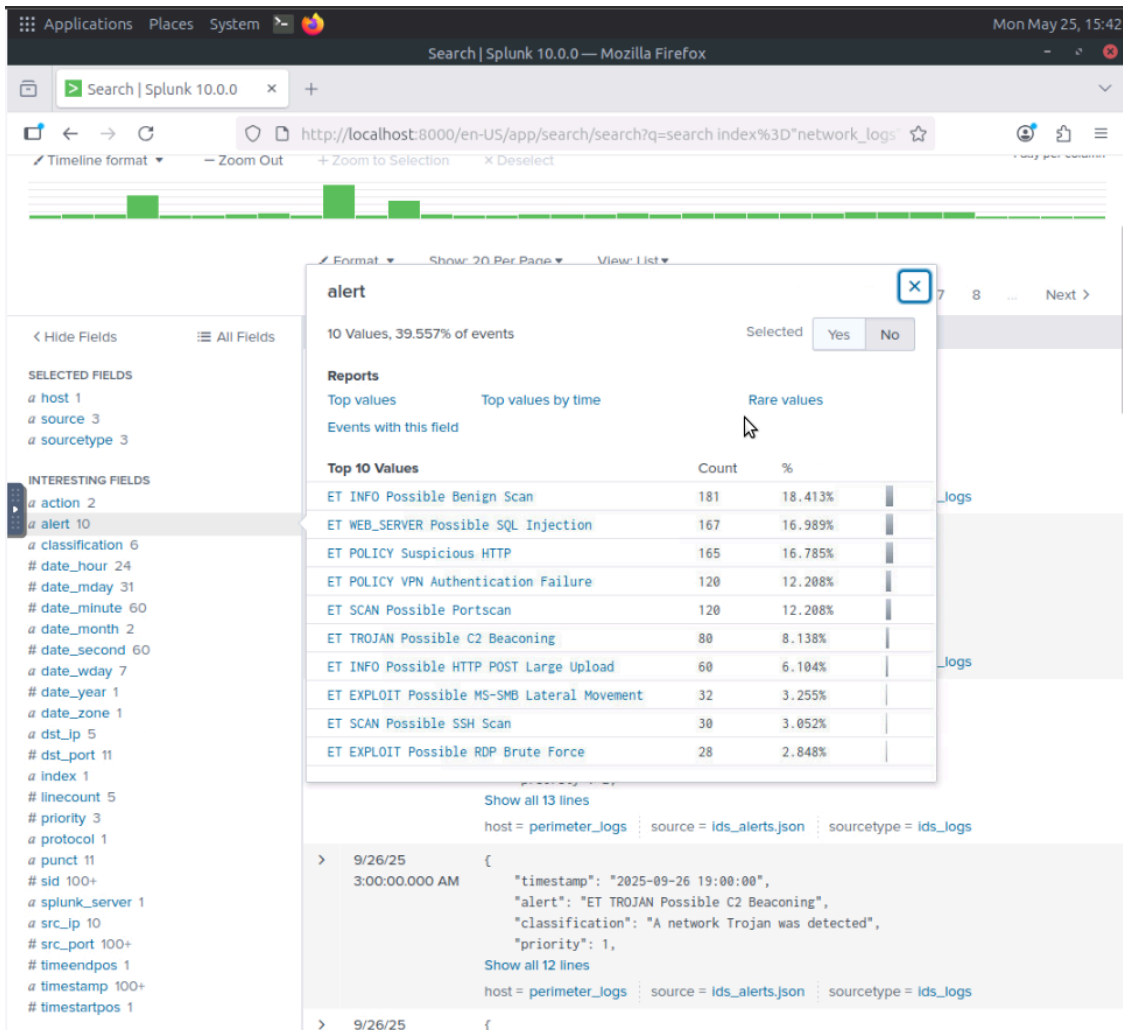


Figure 42: Splunk — Top 10 IDS alert types showing full attack landscape across all log sources

6.3 Firewall Log Event Structure in Splunk

- Expanding individual firewall log events in Splunk revealed the full field structure available for analysis: action, bytes, dst_ip, dst_port, protocol, src_ip, src_port, and timestamp.
- This field structure enables precise filtering and correlation across events.
-

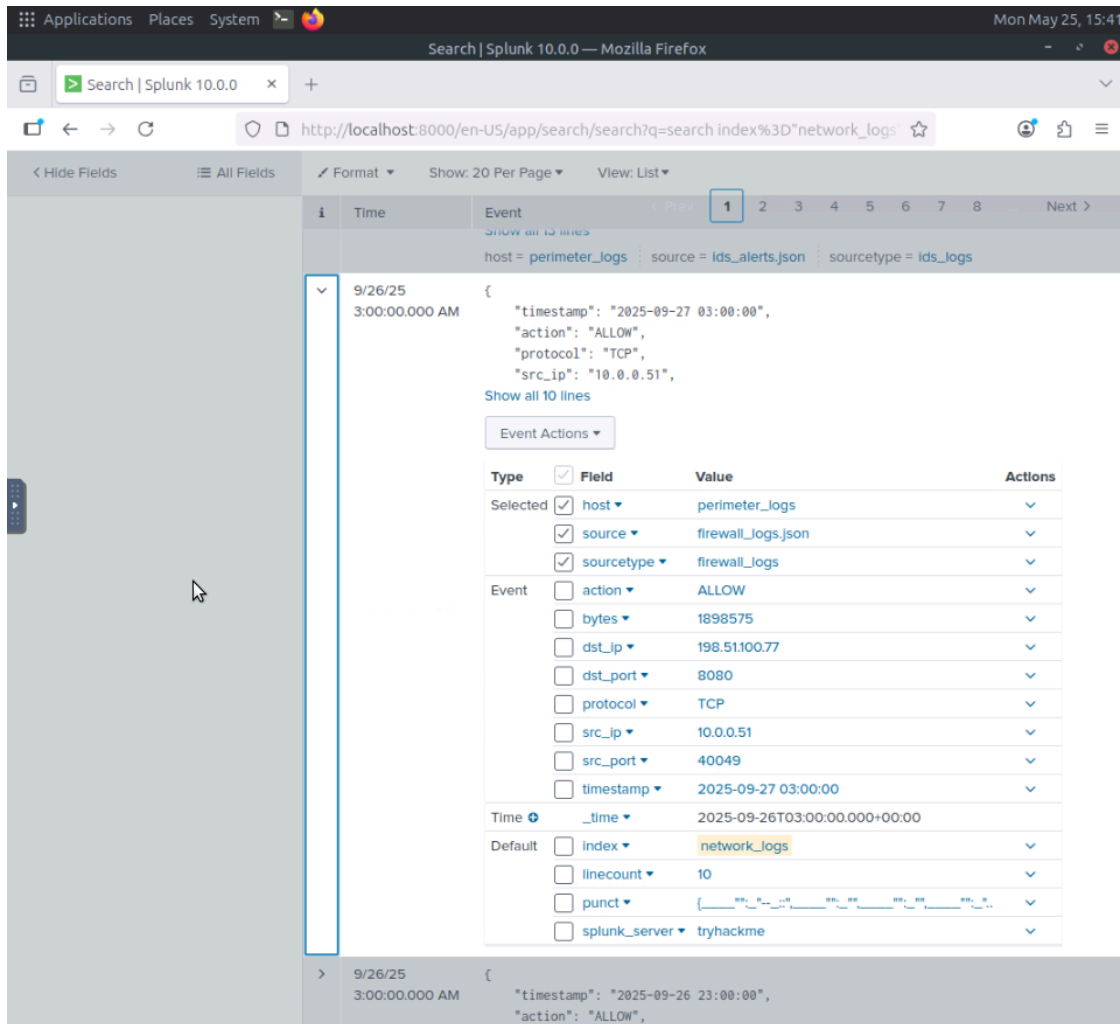


Figure 40: Splunk — expanded firewall log event showing full field structure

6.4 Data Exfiltration Alerts in Splunk

- Splunk IDS alert analysis identified multiple ET INFO Possible HTTP POST Large Upload events classified as Potential Data Exfiltration with Priority 2.
- These events were concentrated on 2025-09-26 and correlated with the C2 beaconing activity observed in the IDS logs, suggesting the compromised host at 10.0.0.60 was actively exfiltrating data.

The screenshot shows the Splunk search interface in a web browser. The search bar at the top contains the query `search index%3Dnetwork_logs`. The left sidebar displays a list of fields, including `host`, `source`, `source_type`, `classification`, `priority`, `timestamp`, and `event`. The main panel shows a list of events with the following details:

Time	Event
9/26/25 3:00:00.000 AM	<pre>{ "classification": "Potential Data Exfiltration", "priority": 2, "timestamp": "2025-09-26 23:00:00", "alert": "ET INFO Possible HTTP POST Large Upload", "classification": "Potential Data Exfiltration", "priority": 2, }</pre> host = perimeter_logs source = ids_alerts.json sourcetype = ids_logs
9/26/25 3:00:00.000 AM	<pre>{ "timestamp": "2025-09-26 19:00:00", "alert": "ET INFO Possible HTTP POST Large Upload", "classification": "Potential Data Exfiltration", "priority": 2, }</pre> host = perimeter_logs source = ids_alerts.json sourcetype = ids_logs
9/26/25 3:00:00.000 AM	<pre>{ "timestamp": "2025-09-26 19:00:00", "alert": "ET TROJAN Possible C2 Beaconing", "classification": "A network Trojan was detected", "priority": 1, }</pre> host = perimeter_logs source = ids_alerts.json sourcetype = ids_logs
9/26/25 3:00:00.000 AM	<pre>{ "timestamp": "2025-09-26 15:00:00", "alert": "ET INFO Possible HTTP POST Large Upload", "classification": "Potential Data Exfiltration", "priority": 2, }</pre> host = perimeter_logs source = ids_alerts.json sourcetype = ids_logs
9/26/25 3:00:00.000 AM	<pre>{ "timestamp": "2025-09-26 13:00:00", }</pre>

Figure 35: Splunk — data exfiltration alerts and C2 beaconing events correlated in SIEM

Applications Places System Search | Splunk 10.0.0 — Mozilla Firefox Mon May 25, 15:39

Search | Splunk 10.0.0

http://localhost:8000/en-US/app/search/search?q=search index%3D"network_logs"

< Hide Fields All Fields Format Show: 20 Per Page View: List You're signed out. Click the account icon to sign in.

1 2 3 4 5 6 7 8 Next >

Time	Event																																																																																
3:00:00.000 AM	<pre>{ "timestamp": "2025-09-26 19:00:00", "alert": "ET INFO Possible HTTP POST Large Upload", "classification": "Potential Data Exfiltration", "priority": 2, }</pre> Show all 13 lines Event Actions <table border="1"> <thead> <tr> <th>Type</th> <th>Field</th> <th>Value</th> <th>Actions</th> </tr> </thead> <tbody> <tr> <td>Selected</td> <td>☒ host</td> <td>perimeter_logs</td> <td></td> </tr> <tr> <td></td> <td>☒ source</td> <td>ids_alerts.json</td> <td></td> </tr> <tr> <td></td> <td>☒ sourcetype</td> <td>ids_logs</td> <td></td> </tr> <tr> <td>Event</td> <td>☐ alert</td> <td>ET INFO Possible HTTP POST Large Upload</td> <td></td> </tr> <tr> <td></td> <td>☐ bytes</td> <td>1891224</td> <td></td> </tr> <tr> <td></td> <td>☐ classification</td> <td>Potential Data Exfiltration</td> <td></td> </tr> <tr> <td></td> <td>☐ dst_ip</td> <td>198.51.100.77</td> <td></td> </tr> <tr> <td></td> <td>☐ dst_port</td> <td>8080</td> <td></td> </tr> <tr> <td></td> <td>☐ priority</td> <td>2</td> <td></td> </tr> <tr> <td></td> <td>☐ protocol</td> <td>TCP</td> <td></td> </tr> <tr> <td></td> <td>☐ sid</td> <td>2002047</td> <td></td> </tr> <tr> <td></td> <td>☐ src_ip</td> <td>10.0.0.51</td> <td></td> </tr> <tr> <td></td> <td>☐ src_port</td> <td>40047</td> <td></td> </tr> <tr> <td></td> <td>☐ timestamp</td> <td>2025-09-26 19:00:00</td> <td></td> </tr> <tr> <td>Time</td> <td>☐ _time</td> <td>2025-09-26T03:00:00.000+00:00</td> <td></td> </tr> <tr> <td>Default</td> <td>☐ index</td> <td>network_logs</td> <td></td> </tr> <tr> <td></td> <td>☐ linecount</td> <td>13</td> <td></td> </tr> <tr> <td></td> <td>☐ punct</td> <td>{ _source _type _id _score }</td> <td></td> </tr> <tr> <td></td> <td>☐ splunk_server</td> <td>tryhackme</td> <td></td> </tr> </tbody> </table>	Type	Field	Value	Actions	Selected	☒ host	perimeter_logs			☒ source	ids_alerts.json			☒ sourcetype	ids_logs		Event	☐ alert	ET INFO Possible HTTP POST Large Upload			☐ bytes	1891224			☐ classification	Potential Data Exfiltration			☐ dst_ip	198.51.100.77			☐ dst_port	8080			☐ priority	2			☐ protocol	TCP			☐ sid	2002047			☐ src_ip	10.0.0.51			☐ src_port	40047			☐ timestamp	2025-09-26 19:00:00		Time	☐ _time	2025-09-26T03:00:00.000+00:00		Default	☐ index	network_logs			☐ linecount	13			☐ punct	{ _source _type _id _score }			☐ splunk_server	tryhackme	
Type	Field	Value	Actions																																																																														
Selected	☒ host	perimeter_logs																																																																															
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Event	☐ alert	ET INFO Possible HTTP POST Large Upload																																																																															
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	☐ punct	{ _source _type _id _score }																																																																															
	☐ splunk_server	tryhackme																																																																															

4 more fields
+ Extract New Fields

9/26/25 {
3:00:00.000 AM {
"timestamp": "2025-09-26 19:00:00",
"alert": "ET TROJAN Possible C2 Beaconing"

Figure 36: Splunk — IDS alert timeline showing attack progression

Applications Places System Search | Splunk 10.0.0 — Mozilla Firefox Mon May 25, 15:39

Search | Splunk 10.0.0 x +

http://localhost:8000/en-US/app/search/search?q=search index%3D*network_logs

< Hide Fields All Fields Format Show: 20 Per Page View: List

priority 3
a protocol 1
a punct 11
sid 100+
a splunk_server 1
a src_ip 10
src_port 100+
timeendpos 1
a timestamp 100+
timestartpos 1

4 more fields
+ Extract New Fields

1 2 3 4 5 6 7 8 Next >

9/26/25 3:00:00.000 AM {
"timestamp": "2025-09-26 19:00:00",
"alert": "ET TROJAN Possible C2 Beaconsing",
"classification": "A network Trojan was detected",
"priority": 1,
Show all 12 lines
Event Actions

Type	Field	Value	Actions
Selected	☒ host	perimeter_logs	▼
	☒ source	ids_alerts.json	▼
	☒ sourcetype	ids_logs	▼
Event	☐ alert	ET TROJAN Possible C2 Beaconsing	▼
	☐ classification	A network Trojan was detected	▼
	☐ dst_ip	198.51.100.77	▼
	☐ dst_port	4444	▼
	☐ priority	1	▼
	☐ protocol	TCP	▼
	☐ sid	2001063	▼
	☐ src_ip	10.0.0.60	▼
	☐ src_port	30063	▼
	☐ timestamp	2025-09-26 19:00:00	▼
Time	☒ _time	2025-09-26T03:00:00.000+00:00	
Default	☐ index	network_logs	▼
	☐ linecount	12	▼
	☐ punct	[...]	▼
	☐ splunk_server	tryhackme	▼

> 9/26/25 3:00:00.000 AM {
"timestamp": "2025-09-26 15:00:00",
"alert": "ET INFO Possible HTTP POST Large Upload",

Figure 37: Splunk — additional SIEM correlation results

Applications Places System  Mon May 25, 15:40

Search | Splunk 10.0.0 — Mozilla Firefox

Search | Splunk 10.0.0

http://localhost:8000/en-US/app/search/search?q=search index%3D*network_logs

< Hide Fields All Fields Format Show: 20 Per Page View: List

timeendpos 1
a timestamp 100+
timestartpos 1

4 more fields
+ Extract New Fields

9/26/25 3:00:00.000 AM

```
{
  "timestamp": "2025-09-26 15:00:00",
  "alert": "ET INFO Possible HTTP POST Large Upload",
  "classification": "Potential Data Exfiltration",
  "priority": 2,

```

Show all 13 lines

Event Actions

Type	Field	Value	Actions
Selected	☒ host	perimeter_logs	v
	☒ source	ids_alerts.json	v
	☒ sourcetype	ids_logs	v
Event	☐ alert	ET INFO Possible HTTP POST Large Upload	v
	☐ bytes	1806851	v
	☐ classification	Potential Data Exfiltration	v
	☐ dst_ip	198.51.100.77	v
	☐ dst_port	8080	v
	☐ priority	2	v
	☐ protocol	TCP	v
	☐ sld	2002046	v
	☐ src_ip	10.0.0.51	v
	☐ src_port	40046	v
	☐ timestamp	2025-09-26 15:00:00	v
Time	☐ _time	2025-09-26T03:00:00.000+00:00	v
Default	☐ index	network_logs	v
	☐ linecount	13	v
	☐ punct	(...)	v
	☐ splunk_server	tryhackme	v

Figure 38: Splunk — Splunk search results showing alert classifications

Applications Places System Search | Splunk 10.0.0 — Mozilla Firefox Mon May 25, 15:4

Search | Splunk 10.0.0

http://localhost:8000/en-US/app/search/search?q=search index%3D"network_logs"

< Hide Fields All Fields Format Show: 20 Per Page View: List

	i	Time	Event	1	2	3	4	5	6	7	8	Next >
	9/26/25	3:00:00.000 AM	{									
			"timestamp": "2025-09-26 07:00:00",									
			"alert": "ET TROJAN Possible C2 Beaconsing",									
			"classification": "A network Trojan was detected",									
			"priority": 1,									
			Show all 12 lines									
			Event Actions									
			Type	☒ Field	Value							Actions
			Selected	☒ host	perimeter_logs							
				☒ source	ids_alerts.json							
				☒ sourcetype	ids_logs							
			Event	☐ alert	ET TROJAN Possible C2 Beaconsing							
				☐ classification	A network Trojan was detected							
				☐ dst_ip	198.51.100.77							
				☐ dst_port	4444							
				☐ priority	1							
				☐ protocol	TCP							
				☐ sid	2001061							
				☐ src_ip	10.0.0.60							
				☐ src_port	30061							
				☐ timestamp	2025-09-26 07:00:00							
			Time	☒ _time	2025-09-26T03:00:00.000+00:00							
			Default	☐ index	network_logs							
				☐ linecount	12							
				☐ punct	[...]							
				☐ splunk_server	tryhackme							
	>	9/26/25	{									
		3:00:00.000 AM	"timestamp": "2025-09-26 03:00:00",									
			"alert": "ET TROJAN Possible C2 Beaconsing",									

Figure 39: Splunk — final SIEM query results

7. Lab Completion

- The TryHackMe Network Security Essentials room was completed in full.
- All 8 tasks were completed, earning 96 points.

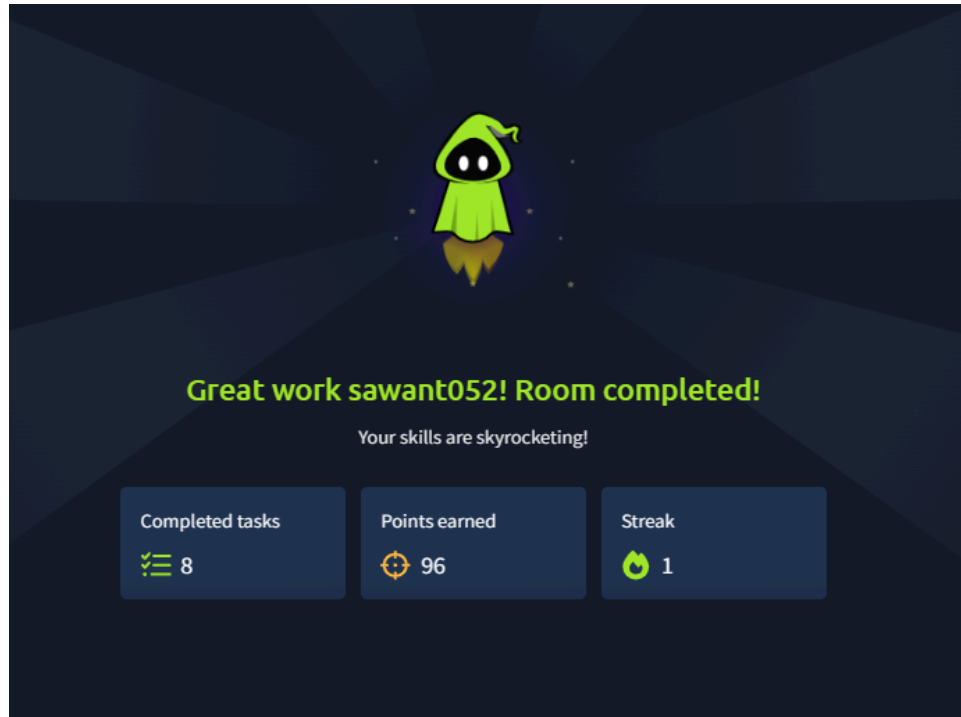


Figure 30: TryHackMe — Room completed: 8 tasks, 96 points, user sawant052

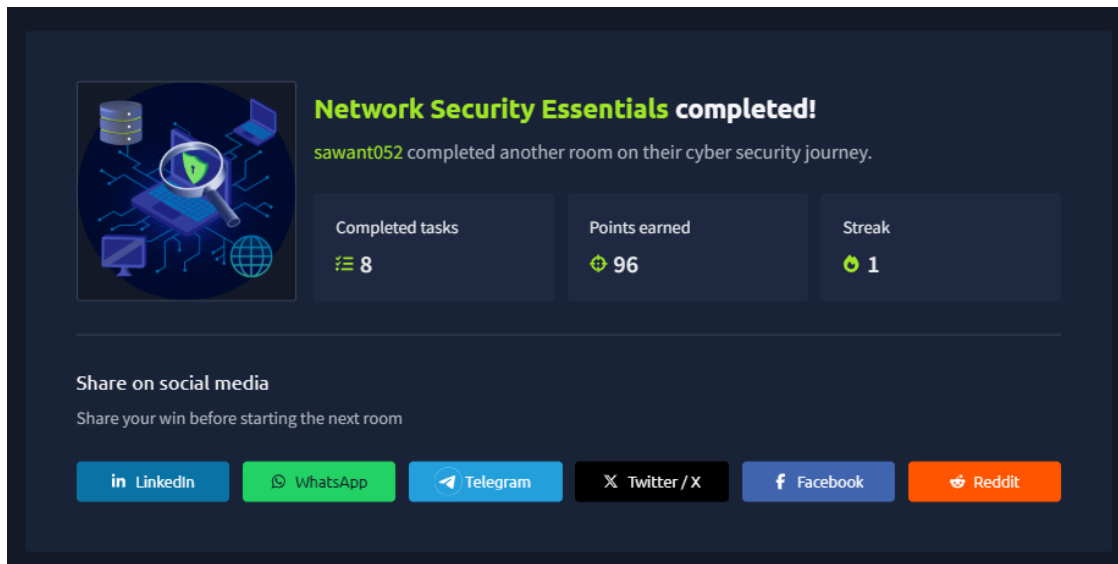


Figure 31: TryHackMe — Network Security Essentials completion certificate

8. Summary of Findings

- The following attack types were identified and analysed during the lab:
- SQL injection
 - WAF rule 942100 blocked UNION SELECT attack from 198.51.100.12 targeting /products.php
- XSS
 - WAF rule 941100 blocked script injection from 198.51.100.12 targeting /search.php
- Directory traversal
 - WAF rule 930120 blocked path traversal attempt to /etc/passwd
- VPN brute-force
 - 45.137.22.13 performed credential stuffing against VPN gateway using 30+ usernames
- Internal port scan
 - host 10.8.0.23 generated high-frequency traffic across ports 22, 445, 3389, 8080
- C2 beaconing
 - host 10.0.0.60 communicated to 198.51.100.7:4444 at 6-hour intervals
- Data exfiltration
 - large HTTP POST uploads from compromised host correlated with C2 activity in Splunk
- MS-SMB lateral movement
 - 32 ET EXPLOIT alerts indicating attempted propagation across internal network
- These findings directly validate the perimeter security controls documented in the Perimeter Security Configuration Guide, confirming that NGFW, WAF, VPN gateway, and IDPS controls generate actionable, correlatable log data that enables effective threat detection and investigation.